

Aircraft Spray Pattern Testing Software
User Manual - Version 2.2

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1 Introduction

AccuPatt is a desktop application created to process spray pattern data for agricultural aircraft. This includes measuring tracer dye deposition on string collectors and/or measuring droplet deposition from spray cards. The goals of this application are twofold:

- Maximize flexibility through parameterization to provide broad utility
- Wherever possible, simplify; The best tools don't need user manuals (And yet here we are)

1.1 Measuring Deposition with String

The use of string for assessing aircraft spray patterns is a longstanding methodology. As a small, continuous collector, it offers both high collection efficiency for smaller droplets and high spatial resolution across a swath. The tradeoffs include a lower collection efficiency for larger droplets and its limit to measuring only relative deposition.

By using a fluorescent tracer dye in the spray solution, the relative deposition along the string can be measured by exciting the dye with a light source of the appropriate wavelength and measuring the intensity of emitted light at another wavelength.

This method is particularly useful for rapidly assessing pattern shape and uniformity, but it lacks the ability to provide quantifiable deposition data. It also offers no insight into droplet size distribution.

1.2 Measuring Deposition with Spray Cards

Spray cards (or paper) are another method for assessing aircraft spray patterns. By capturing droplet stains on a flat surface, they can be scanned and processed to provide quantifiable deposition data as well as insight into droplet size distribution. The tradeoffs include lower spatial resolution across the swath (unless continuous paper is used) and lower collection efficiency for smaller droplets.

Spray cards can be used as a standalone method for assessing spray patterns, but they are often used in conjunction with string to provide a more complete picture of the pattern. For example, string can be used to rapidly assess pattern shape and uniformity while spray cards can be used to provide quantifiable deposition data and insight into droplet size distribution.

There are three basic types of paper typically used for pattern testing:

- Water-Sensitive Paper (WSP) is a discrete sampler that changes color when it comes into contact with water; this allows it to be used without the need for a dye in the spray solution.
- Cast-Coat Paper (Kromekote, etc.) is a discrete sampler that has a uniform, glossy finish which permits acquisition of smaller stains than WSP when scanned at higher resolutions. However, it requires the use of a dye in the spray solution for the stains to be visible.
- Bond Paper (Receipt Paper, etc.) is a continuous sampler (available in rolls) that can be deployed rapidly. It also requires the use of a dye in the spray solution for the stains to be visible.

The major caveat to all spray card/paper analysis is that, while droplet density and relative coverage can be measured, the translation of stain size to droplet size is not straightforward. This is because the stain size is a function of both the droplet size and the spread factor, which is influenced by various factors such as spray solution properties, humidity, and paper properties. Therefore, while spray card analysis can provide insight into droplet size distribution, it should be interpreted with extreme caution and an understanding of its limitations.

2 Compatible Hardware

2.1 String Hardware

AccuPatt was designed around the *WRK of Oklahoma* String Spectrometer System, and thus is fully compatible with that system. It uses a combination wavelength-specific LED and Band-Pass filter as the excitation source and a commercial USB Spectrometer (Ocean Optics, USB2000+, Flame or similar) as the emission wavelength measurement device. It utilizes a set of serially controlled stepper motors in conjunction with various clutches and pulleys to move the string at a constant speed. Any questions on the *WRK of Oklahoma* system should be directed to its creator, Dr. Richard Whitney (whitney3451@att.net)

2.2 Spray-Card Hardware

To process spray-cards, they must first be scanned with a flatbed scanner. It is recommended to use a dedicated scanner (not a combination printer/scanner) for the highest quality. A supported resolution of at least 1200 dpi is recommended for most applications. *AccuPatt* can communicate with the scanner directly (MacOS Only) or images can be scanned externally using the software of your choice and uploaded to *AccuPatt* for processing.

3 Installation

3.1 Windows

Windows 10 or newer is required.

1. Download and install the *OceanDirect* spectrometer driver [provided by the manufacturer here](#).
2. Download and install the *FTDI (VCP)* Stepper Motor driver [provided by the manufacturer here](#).
3. Download and install *AccuPatt* from [its GitHub repository here](#). (You just need the *Install.AccuPatt.exe* file)

3.2 MacOS

MacOS 11 or newer, running on ARM64 (Apple Silicon), required.

1. Download *AccuPatt* from [its GitHub repository here](#) (You just need the *AccuPatt.dmg* file).
2. Install *AccuPatt* by opening the downloaded .dmg as a volume. Drag the *AccuPatt* icon to the *Applications* shortcut icon. Since this app isn't registered with Apple, you will need to right click on the app and choose *Open*. You can find more detail on this process from [Apple's support page](#).

4 Settings & Configuration

Choosing **File** → **Settings** from the menu will open the **Settings** window, which contains several tabs for configuring various aspects of the application. These are saved persistently and applied to all newly created series. Some settings, such as plot preferences which do not affect pattern/data processing, will also apply to previously created series which are opened subsequently.

4.1 General Settings

- **Data File Directory:** This is the directory which data files will be saved into by default, as well as where the application will look for data files when using the **New Series** → **File Aircraft** option.
- **Fly-In Default Values:** The Name, Location, Date and Analyst(s) fields automatically applied to new series.
- **Passes:** The number of passes to be used in new series.

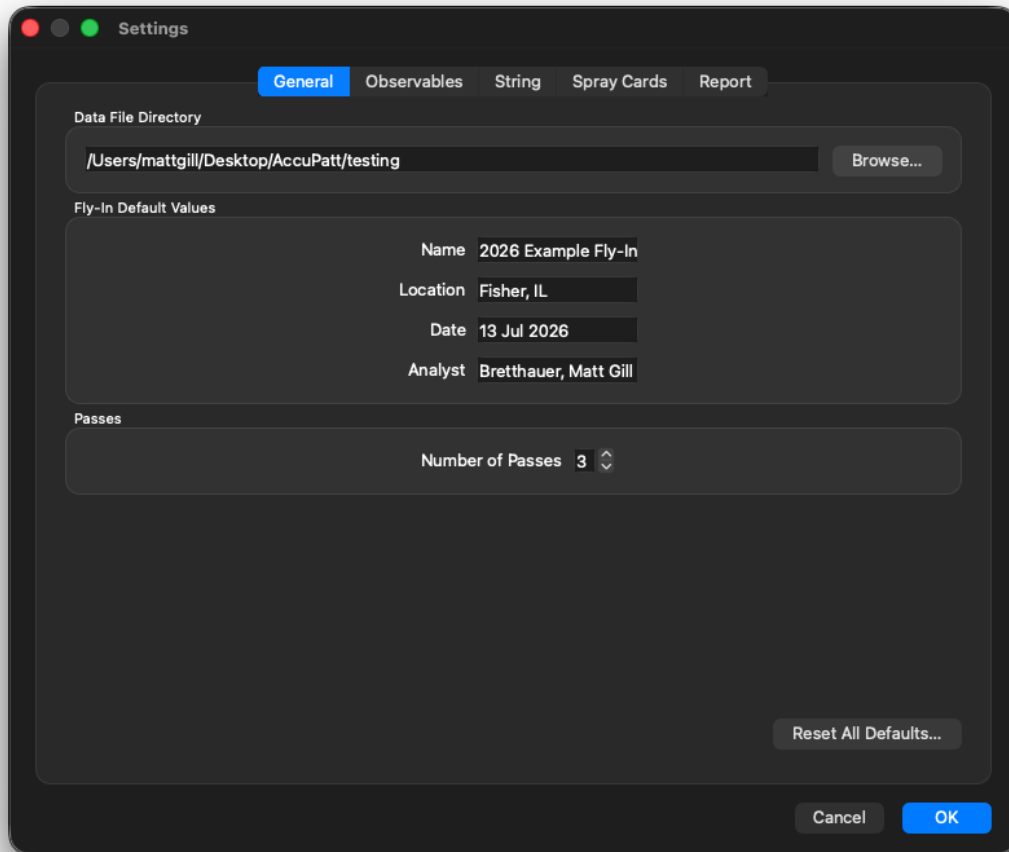


Figure 1: Settings - General Tab

4.2 Observables Settings

- **Series Observables:** The default units for new series observables.
- **Pass Observables:** The default units for pass observables in a new series.

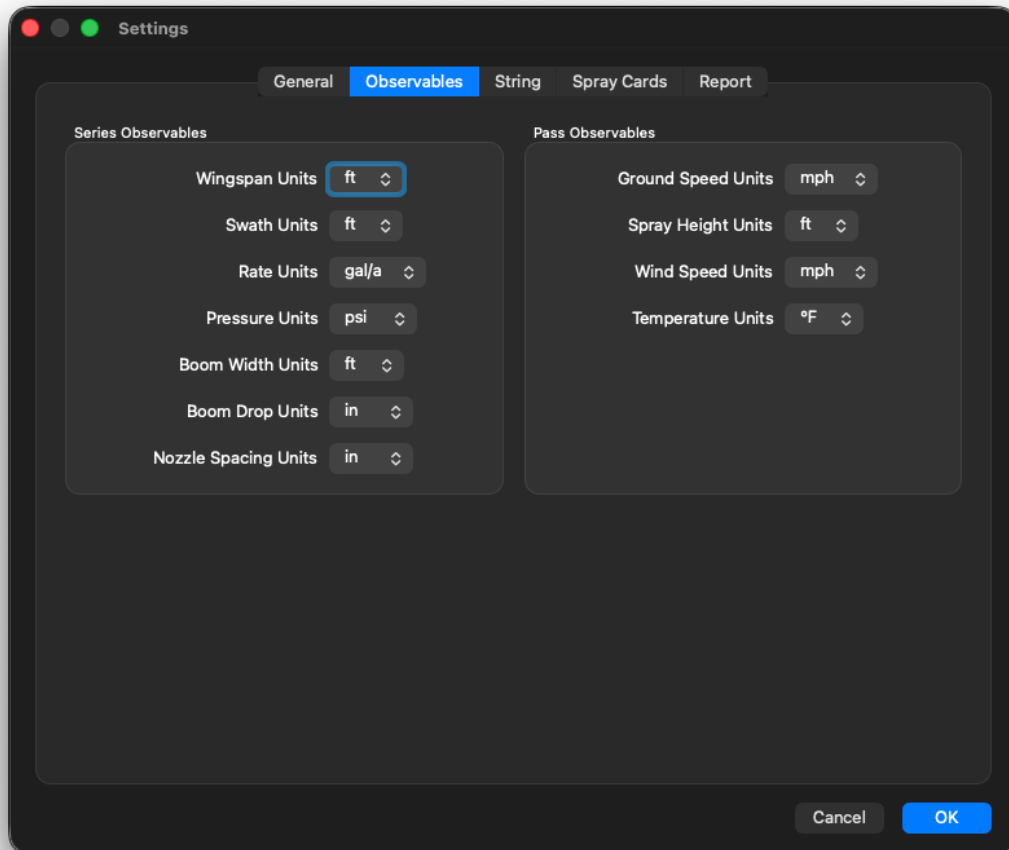


Figure 2: Settings - Observables Tab

4.3 String Settings

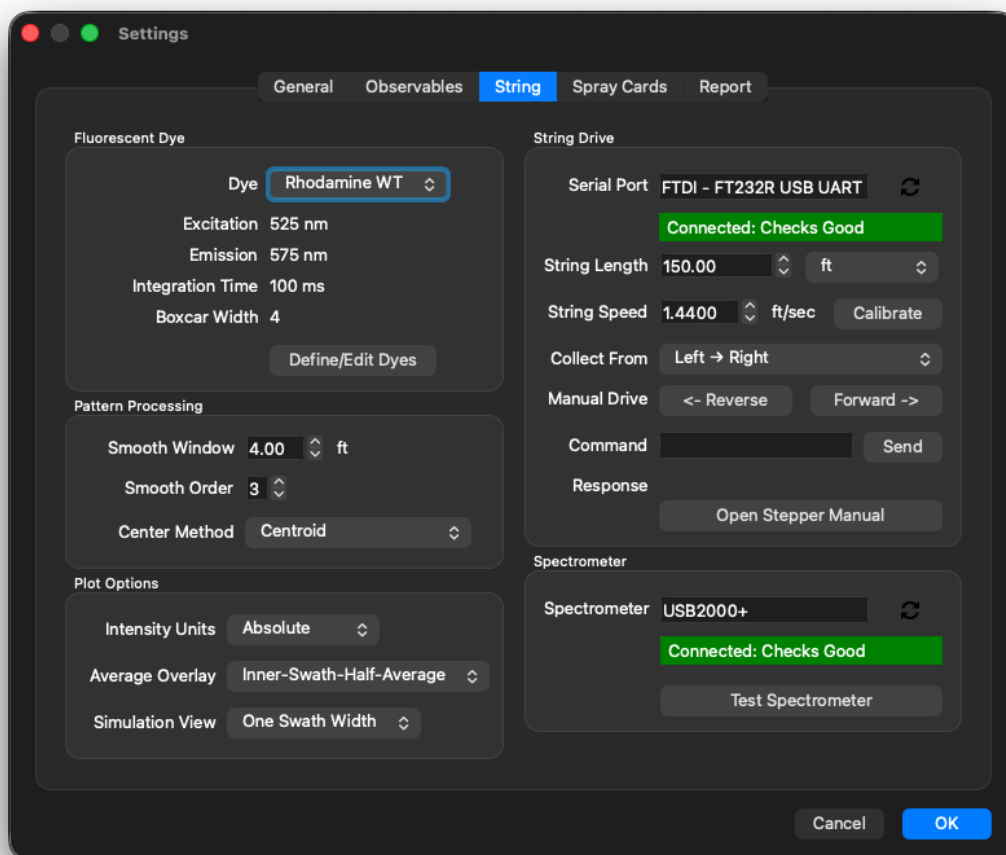


Figure 3: Settings - String Tab

4.3.1 Fluorescent Dye

The default dye for new series. The listed parameters associated with the dye are used for configuring and acquiring data from the spectrometer. Several pre-loaded dyes are included, but new dyes can be added via the **Define/Edit Dyes** button which opens an edit window (Figure 4).

The dye parameters are as follows:

- **Excitation Wavelength (nm):** The target wavelength of the excitation light source (LED + Band-Pass filter). Changing this value will NOT automatically adjust the excitation source; the proper LED and Band-Pass filter must be physically installed. While excitation data is stored in the datafile, *AccuPatt* only utilizes it as a reference point in the **Test Spectrometer** window.
- **Emission Wavelength (nm):** The target wavelength of light emitted by the dye when excited. This is the wavelength at which intensity is captured as the relative dye deposition.

- **Integration Time (ms):** Time to collect light for a spectrometer reading. Think of this as the exposure on a camera; longer times correspond with more detail incorporated in each data frame, but a lower quantity of data frames per unit length of string.
- **Boxcar Width (nm):** The cross-spectrum wavelength width to average for each spectrometer reading. A value of 0 means to take a reading only from the spectrometer pixel nearest the target emission wavelength. A value of 5, for example, would have each spectrometer reading be an average of every pixel within 5 nm (± 2.5 nm) of the target emission wavelength.

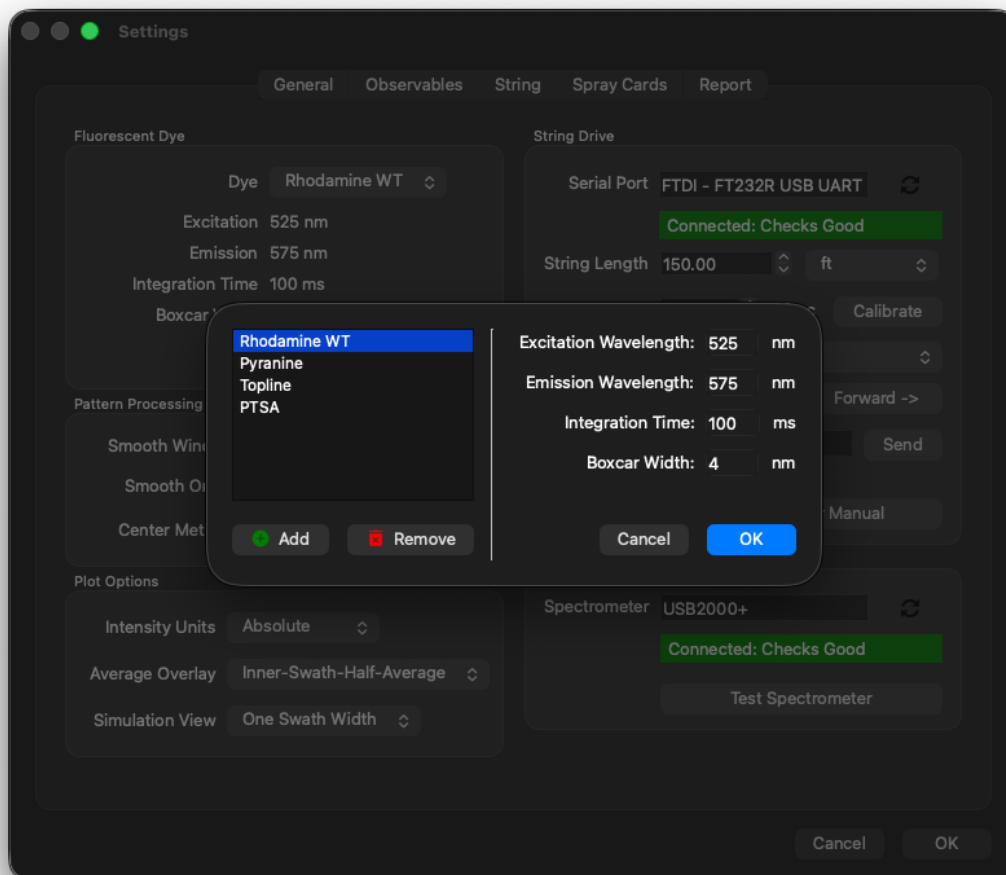


Figure 4: Settings - String Tab - Define/Edit Dyes

4.3.2 Pattern Processing

The default smoothing and centering applied to new series; both for individual passes and the average pattern. These settings can be adjusted on a per-pass or per-series basis from the **String Analysis** tab as well.

- **Smooth Window:** X-domain based distance over which the sliding smoothing filter is fit and applied

- **Smooth Order:** Polynomial order for smoothing filter
- **Center Method:** The pattern is horizontally shifted so either the centroid or center of distribution is the new zero. (These are equivalent for evenly spaced data, so it typically doesn't matter for regularly spaced string data)

4.3.3 Plot Options

The visual-only settings for string plots; these are not saved with datafiles and do not affect computations of CV.

- **Intensity Units:** The y-axis units for emission data intensity. These are always stored in Absolute Units (AU) but may optionally be displayed as a Relative (%) of the spectrometer's 16-bit saturation value (65535 AU).
- **Average Overlay:** The dashed line optionally displayed as an overlay to the average pattern as one of:
 - **Inner-Swath-Half-Average:** Computes one-half the average y-value within the **Adjusted Swath Width** range and draws a horizontal line at that value bounded by vertical lines at the bounds of the **Adjusted Swath Width**. This is sometimes referred to as the "old-school" way of determining effective swath width, where the intersection points of the horizontal and vertical lines should approximately touch the plotted pattern.
 - **Average:** Computes the average y-value of the entire pattern and draws a horizontal line at that value.
- **Simulation View:** Set the x-bounds of the overlap simulation plots as one of:
 - **One Swath Width:** Only the mathematically relevant portion of the simulated overlap will be shown, specifically datapoints between $-\text{swathwidth}/2$ and $\text{swathwidth}/2$.
 - **All Swaths:** All selected simulated adjacent passes will be shown. Can be useful for illustrating the big picture of in-field overlap.

4.3.4 String Drive

The settings for controlling the stepper motors in capturing string data; not saved to the datafile.

- **Serial Port:** Searches for and automatically connects to an FTDI device if found. A refresh button allows a manual connection attempt. A colored status readout shows green if the device is ready and yellow if it is not.
- **String Length:** The length and units (ft or m) of string to process for a single pass.
- **String Speed:** The calibrated velocity of the string as it is processed. This is used to determine the location of each data point and duration of data capture for each pass. A **Calibrate** button opens an interactive calibration window ([Figure 5](#)) to use a known length of string (defaults to flightline length but any length may be used) to calculate the speed.
- **Collect From:** The orientation Left to Right or Right to Left in which the data should be collected. This is typically Left to Right (unwinding string spools previously collected in the field) but should be set as Right to Left if processing string directly in the field from the aircraft's right side.
- **Manual Drive:** Buttons to activate the forward or reverse motors for testing or ad-hoc needs.

- **Command:** Facilitates direct querying and setting string drive stepper motor parameters to onboard nonvolatile memory. The Take-Up Reel corresponds to address **A** and the Supply Reel corresponds to address **B**. With a command typed into the text field, clicking the **Send** button will append a carriage return, encode the message and send it to the stepper motor driver. A string of return text is normally received, and is displayed when available. This return text may simply be an echo of your command, or it may contain parameter information. See the stepper motor driver documentation by clicking the **Open Stepper Manual** button for specific commands.

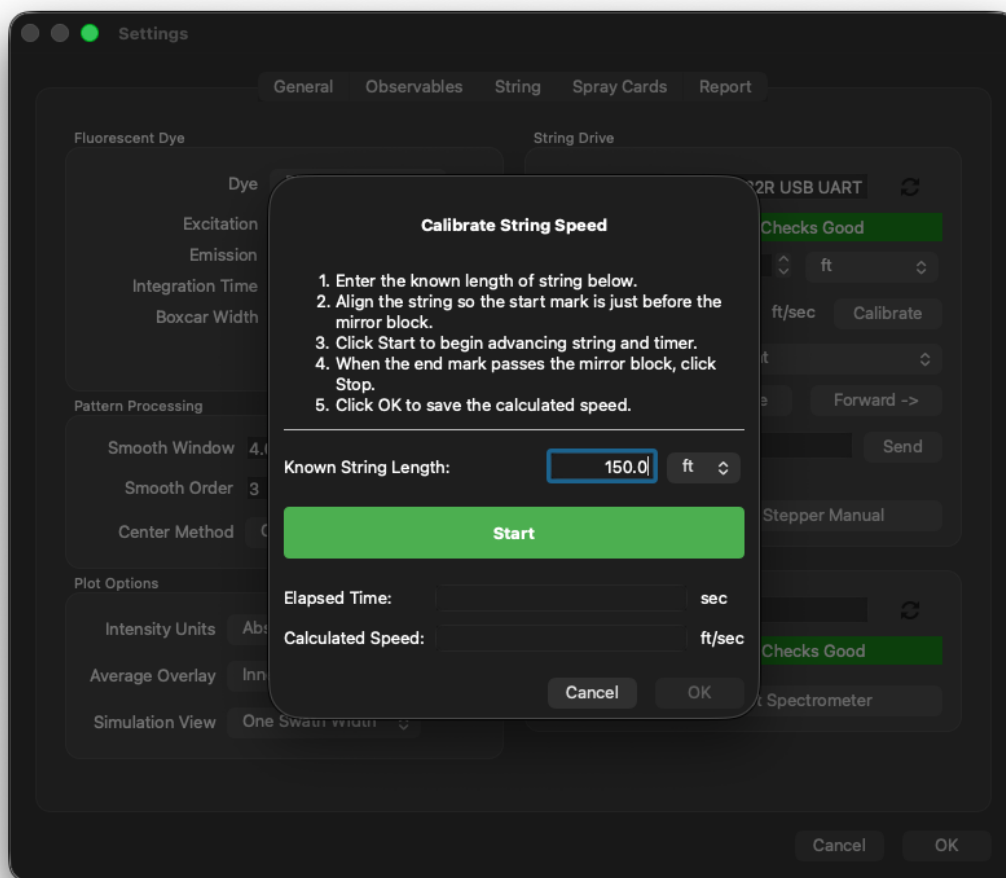


Figure 5: Settings - String Tab - Calibrate Speed

4.3.5 Spectrometer

An automatic search and connection (if available) for a connected spectrometer. A refresh button allows for a manual connection attempt. A colored status readout shows green if the device is ready, yellow if it is not and orange if there is a driver issue.

The **Test Spectrometer** button opens a window with a live readout of the entire measured spectrum (Figure 6). The previously selected dye is used to set target excitation and emission wavelengths as

well as integration time and boxcar width.

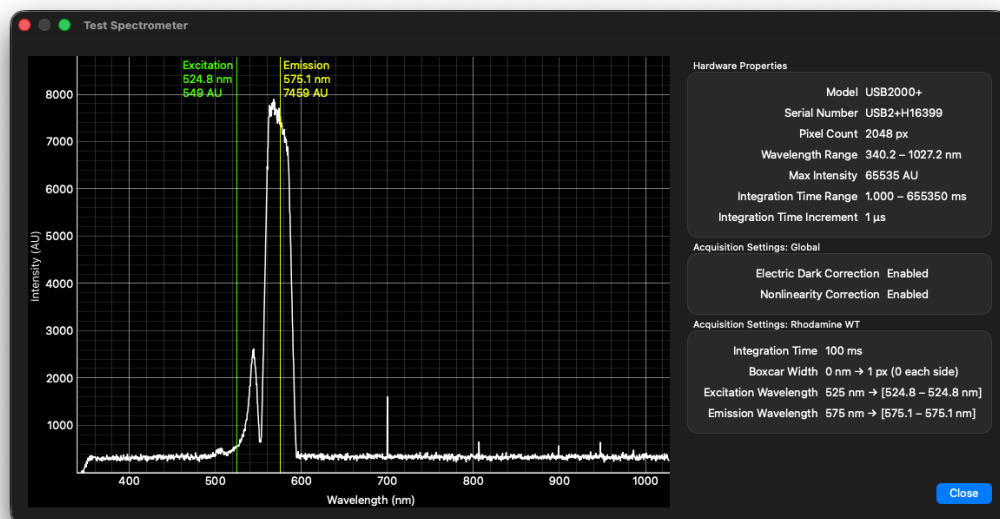


Figure 6: Settings - String Tab - Test Spectrometer (With Emission Filter)

Some installations do not include an emission filter. In this case the plot will show the excitation source maxing out the scale and an emission "hump" where data is measured. Figure 7 is provided below to show what that should look like for a typical Rhodamine setup with a 525nm band-pass excitation filter and no emission filter.

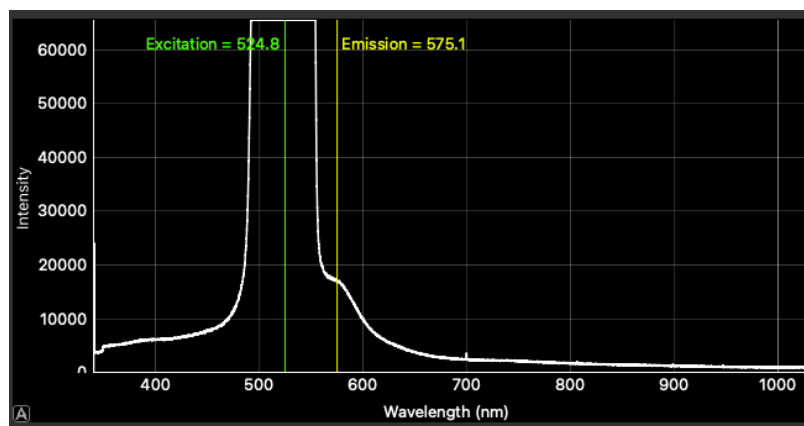


Figure 7: Settings - String Tab - Test Spectrometer (No Emission Filter)

4.4 Spray Cards

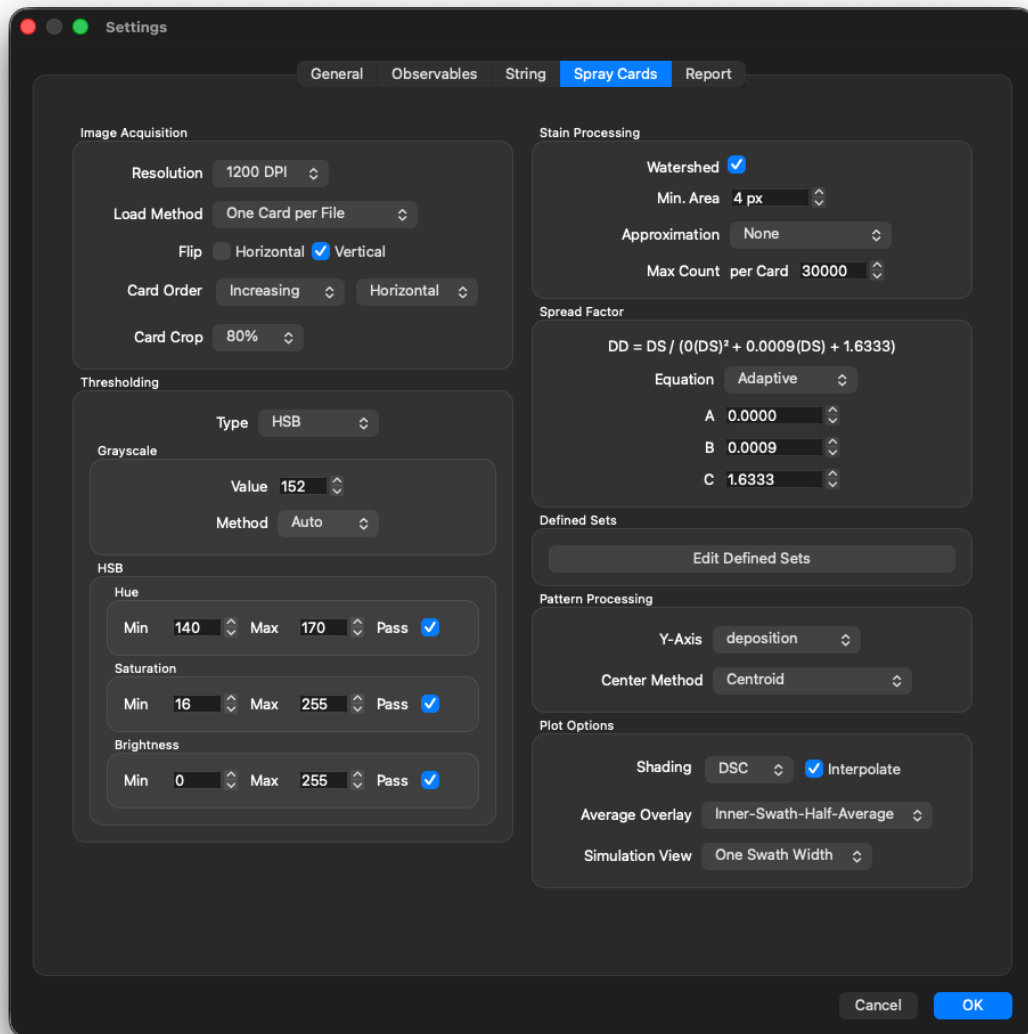


Figure 8: Settings - Spray Cards Tab

4.4.1 Image Acquisition

These settings are persistent but are not saved to the datafile since they do not affect the processing of the spray card data itself.

Acquiring spray card images to save to the datafile can be done via uploading externally scanned images or through direct scanning in *AccuPatt* (MacOS Only).

- **Load Method:** If scanned externally, select the method to load these images into *AccuPatt* as either:

- **One Card per File:** Each card is externally scanned and saved as a separate image file. When loading, the user will be prompted to select each individual image file for each card. No flipping, cropping or other image manipulations will be performed.
- **Multiple Cards per File:** Multiple cards are externally scanned together in a single image file. When loading, the user will be prompted to select the single image file. The user will then be able to crop each card from the image, with initial automatic selection determined by:
 - * **Flip:** Whether to flip the image horizontally and/or vertically as the first step of automatic selection. This accommodates the various image orientations returned by different scanners and scanning software.
 - * **Card Order:** The order in which cards are arranged in the image file, which determines the initial automatic selection of each card. For example, if there are 4 cards scanned together in a 2x2 grid, and the order is set to *Increasing* and *Horizontal*, then the initial automatic selection for card 1 will be the top left quadrant of the image, card 2 will be the top right quadrant, card 3 will be the bottom left quadrant and card 4 will be the bottom right quadrant.
 - * **Card Crop:** The percentage of the automatically detected card width and height to crop. While it is best to capture as much of the card as possible, there are sometimes edge defects or slight misalignments in card placement that make it more practicable to set the crop to a value less than 100%.
- **From ZIP:** Multiple cards are externally scanned and saved individually into a ZIP archive file. When loading, the user will be prompted to select the ZIP file. Each card image will be extracted and loaded into *AccuPatt* without any flipping, cropping or other image manipulations. This option is targeted primarily for use with Swath Gobbler™ data sets, which are provided in this format.

4.4.2 Thresholding

These settings are persistent but apply only to new cards; Defined Sets have their own threshold settings on a per-card basis. Moreover, every individual card has its own threshold method and values which will override these default settings for that card. These settings are saved to the datafile on a per-card basis.

Scanned/Loaded card images must be converted to a binary format to distinguish stain pixels from background pixels. Thresholds are set to make this determination for each pixel of the image. *AccuPatt* provides two threshold methods as follows:

- **Grayscale:** The image is converted to 8-bit grayscale (0-255 where 0 is white and 255 is black) and a threshold is set based on pixel intensity. This is the more traditional method for processing spray cards, and is typically more effective for cards with darker stains such as WSP.
 - **Value:** The minimum intensity value for a pixel to be considered a stain pixel. This is typically set based on the intensity histogram of the image, where the goal is to set the threshold at a value that effectively distinguishes the stain pixels from the background pixels.
 - **Method:** The method for determining the threshold value. The **Manual** method uses the user-defined value as the threshold. The **Auto** method uses an algorithm (Otsu's Method) to automatically determine the optimal threshold value based on the intensity histogram of the image.

- **HSB:** The image is converted to Hue, Saturation and Brightness (HSB) color space and a threshold is set based on the Hue, Saturation and Brightness values (8-bit, 0-255) of each pixel. This method can be more effective for cards with lighter stains such as Cast-Coat, where the Hue and Saturation values can be more effective at distinguishing stains from the background than pixel intensity alone. Minimum and Maximum bands can be set for each of the three HSB parameters. If **Pass** is selected for the parameter, a pixel must fall within the defined range for that parameter to be considered a stain pixel (i.e. a band-pass filter is set); if **Pass** is NOT selected, a pixel must fall outside the defined range for that parameter to be considered a stain pixel (i.e. a band-reject filter is set). Note that all three parameters are applied simultaneously, so a pixel must meet the criteria for all three parameters to be considered a stain pixel.

4.4.3 Stain Processing

These settings are persistent but apply only to new cards; Defined Sets have their own stain processing settings on a per-card basis. Moreover, every individual card has its own stain processing settings which will override these default settings for that card. These settings are saved to the datafile on a per-card basis.

Once a card image is thresholded to a binary format, the resulting stain pixels are processed to determine the location and size of each individual stain. This is done through a connected components analysis, where adjacent stain pixels are grouped together into individual stains. The parameters for this process are as follows:

- **Watershed:** Whether to apply a watershed algorithm to separate adjacent stains that may be touching or slightly overlapping. *AccuPatt* uses a marker-controlled watershed algorithm, where the distance transform of the binary image is used to identify the centers of stains as markers, and the watershed algorithm is applied to the distance transform to separate adjacent stains. *It is generally NOT recommended to utilize the watershed algorithm for pattern testing data, as it can introduce more errors than it resolves. Visual verification (via the Outline image where stain borders are drawn on the original image) should always be performed if this option is used.*
- **Min Area:** The minimum area (in pixels) for a group of connected stain pixels to be considered a valid stain. This is typically set to exclude small specks of noise that may be misidentified as stains.
- **Approximation:** Whether to apply a contour approximation algorithm to the detected stains. This will alter stain size and should be used with caution. There approximation methods are as follows:
 - **None:** No contour approximation is applied; the stain size is determined solely by the number of connected stain pixels.
 - **Enclosing Circle:** The stain is approximated as a circle, where the radius of the circle is determined by the distance from the center of the stain to the farthest connected stain pixel. This method can be effective for stains that are roughly circular in shape, but can overestimate the size of irregularly shaped stains.
 - **Fit Ellipse:** The stain is approximated as an ellipse, where the major and minor axes of the ellipse are determined by fitting an ellipse to the connected stain pixels. This method can be effective for stains that are elliptical in shape, but can also overestimate the size of irregularly shaped stains.
 - **Convex Hull:** The stain is approximated as a convex hull, which is the smallest convex

shape that can enclose all of the connected stain pixels. This method can be effective for stains of all shapes, but can overestimate the size of stains with concave features.

- **Max Count:** The maximum number of stains to be detected on a single card. If the number of detected stains exceeds this value, *AccuPatt* will cease processing the image. This is essentially a computational fail-safe for improperly thresholded images that are returning an excessive number of stains, which can cause the application to freeze or crash.

4.4.4 Spread Factor

These settings are persistent but apply only to new cards; Defined Sets have their own spread factor settings on a per-card basis. Moreover, every individual card has its own spread factor settings which will override these default settings for that card. These settings are saved to the datafile on a per-card basis.

Determination of originating droplet size from stain size is not straightforward, as the stain size is a function of both the droplet size and the spread factor, which is influenced by various factors such as spray solution properties, humidity, and paper properties. However, if a spread factor can be determined for the specific conditions of the pattern testing, it can be applied to the stain size to provide an estimate of droplet size.

AccuPatt provides a simplistic approach based upon longstanding industry practice wherein the stain diameter (D_S) is determined from a circle of equal projection area of the stain and the originating droplet diameter (D_D) is determined via one of the following equations:

- **None:** $D_D = D_S$
- **Adaptive:** $D_D = D_S / (aD_S^2 + bD_S + cD_S)$
- **Direct:** $D_D = aD_S^2 + bD_S + cD_S$
- **Logarithmic:** $D_D = a \ln(D_S) + bD_S + c$

Where a , b and c are user-defined coefficients. The **Adaptive** method is the most commonly used, where the stain diameter is divided by the spread factor to determine droplet diameter; for context, this was the method used by *DropletScan*. The **Direct** method has previous documented use in *DepositScan*. The **Logarithmic** method is used in *DropFlight*. The coefficients for the various methods can be found in the associated application's literature or can be determined experimentally for the specific conditions of the pattern testing.

It is important to note that these equations are simplifications and may not accurately capture the complex relationship between droplet size, stain size and spread factor under all conditions. Therefore, any estimates of droplet size derived from stain size should be interpreted with caution and an understanding of the limitations of this approach.

4.4.5 Defined Sets

Defined Sets are persistent.

These are user-defined sets of spray cards, including (for each card) the name, location, threshold settings, stain processing settings and spread factor settings. These are intended to facilitate the processing of pattern testing data by allowing users to save and apply specific settings for different types of cards or testing conditions. For example, a user may have one defined set for processing 9 WSP cards at 8 foot spacing (traditional pattern testing alongside string) with specific threshold, stain processing and spread factor settings, and another defined set for processing 75 Cast-Coat cards at 2 foot spacing with different settings.

Clicking **Edit Defined Sets** will open a window where these sets can be created, edited and deleted (Figure 9). Clicking **Add Set** will insert it into the list of locally defined card sets; double clicking the auto-generated title will allow editing of the name. Individual cards can be added to the set with the **Add Card** button, which appends a card to the set with the default threshold, stain processing and spread factor settings. These can all be edited on a per-card basis by double clicking the applicable fields in the table.

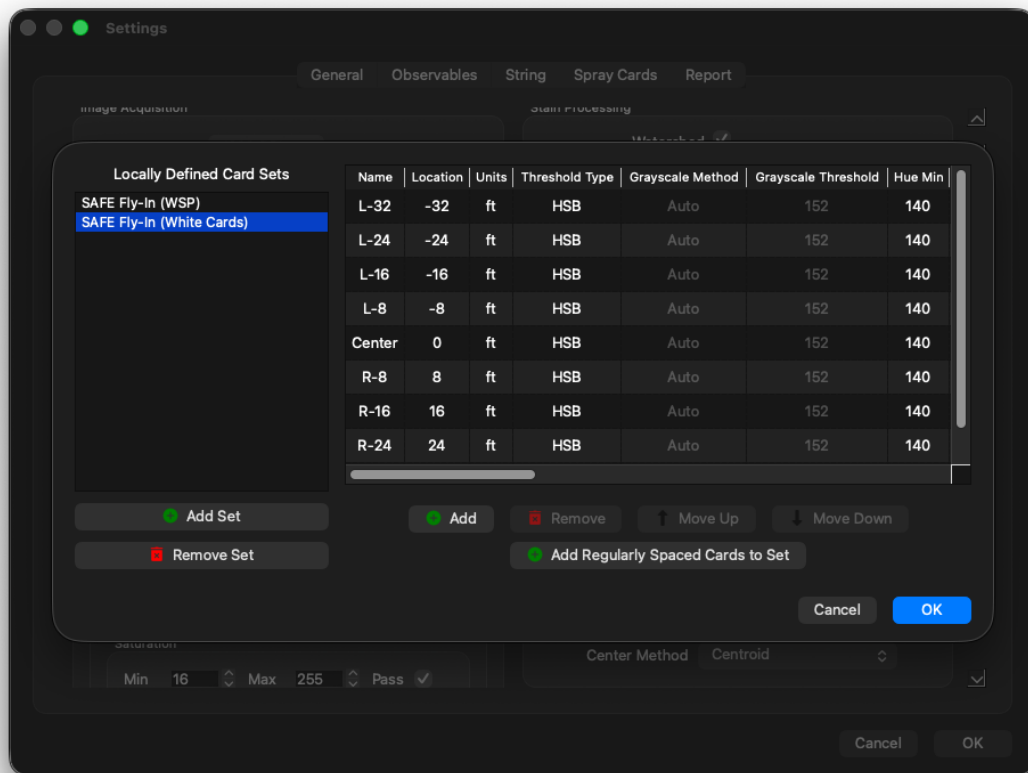


Figure 9: Settings - Spray Cards - Defined Sets

Clicking **Add Regularly Spaced Cards to Set** will open a window where an initial location, quantity of cards and either a step (spacing) or final location can be defined to automatically generate a set of regularly spaced cards (Figure 10). For example, if the initial location is set to 0 feet, the quantity is set to 5 and the step is set to 10 feet, then 5 cards will be generated at locations 0, 10, 20, 30 and 40 feet. Alternatively, if the initial location is set to 0 feet, the quantity is set to 5 and the final location is set to 40 feet, then 5 cards will be generated at locations 0, 10, 20, 30 and 40 feet.

Before clicking **OK** to create the generate the regularly spaced cards, clicking **Select Process Options** will open a window where the threshold settings and stain processing settings for all of the generated cards can be defined. Similarly, clicking **Select Spread Factors** will open a window where the spread factor method and coefficients for all of the generated cards can be defined.

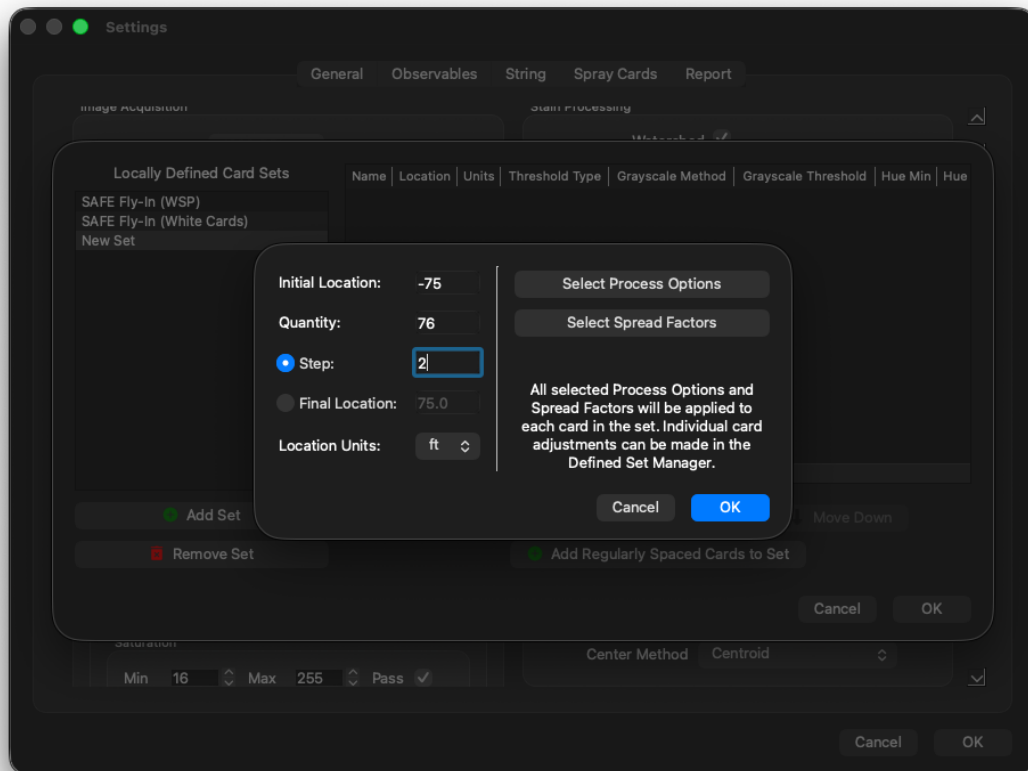


Figure 10: Settings - Spray Cards - Defined Set - Add Regularly Spaced Cards to Set

4.4.6 Pattern Processing

The settings are persistent and apply to new spray card Passes and Series, but can be overridden on a per-Pass or per-Series basis. Only Center Method is saved to the datafile.

These settings affect the processing of spray card data for Pass patterns, Series average pattern and CV calculations. The parameters are as follows:

- **Y-Axis:** The y-axis units for all spray card plots and the basis for swath/uniformity calculations. Available options are:
 - **Coverage:** The percent of the card area covered by stain pixels. This calculation is independent of the spread factor and is therefore the most direct measure of the pattern as captured on the card itself.
 - **Deposition:** The calculated volume per unit area of spray solution deposited on the card, which is derived from the stain size and spread factor. This calculation is dependent on the spread factor and is therefore a more indirect measure of the pattern, but it can be more informative for understanding droplet size distribution and potential efficacy of the application. The units will be either gal/acre or l/ha depending on the Rate Units selected in Series Observables Settings.

- **Drops per cm^2 :** The number of detected stains per unit area of the card. This calculation is independent of the spread factor and is therefore a direct measure of the droplet density on the card, but it does not account for the size of the stains or droplets.
- **Drops per in^2 :** The number of detected stains per unit area of the card. This calculation is independent of the spread factor and is therefore a direct measure of the droplet density on the card, but it does not account for the size of the stains or droplets.
- **Center Method:** The method for determining the center of the pattern, which is used as the reference point to horizontally shift (center) a Pass pattern and Series average pattern for swath and uniformity calculations. The options are:
 - **Centroid:** The x value (location) of the geometric center of the pattern. This is equivalent to the center of distribution (mass) if the space between all cards is uniform.
 - **Center of Distribution:** The x value (location) where the pattern is balanced on either side via weighted averaging. This is useful for patterns where the space between cards is not uniform, as it accounts for the varying distances between cards in determining the center of the pattern.

4.4.7 Plot Options

These settings are persistent but are not saved to the datafile since they do not affect computations of CV.

The visual-only settings for spray card plots.

- **Shading:** Whether to shade the area under the pattern curve on Pass and Series average pattern plots. The only option is **DSC**, which stands for Droplet Spectrum Category, and will use ASABE-standard colors for *Medium*, *Course*, etc. for each card. The shading for each card will extend half the distance to neighboring cards on either side unless **Interpolate** is checked, in which case the shading will be interpolated between cards to create a continuous gradient.
- **Average Overlay:** The dashed line optionally displayed as an overlay to the average pattern as one of:
 - **Inner-Swath-Half-Average:** Computes one-half the average y-value within the **Adjusted Swath Width** range and draws a horizontal line at that value bounded by vertical lines at the bounds of the **Adjusted Swath Width**. This is sometimes referred to as the "old-school" way of determining effective swath width, where the intersection points of the horizontal and vertical lines should approximately touch the plotted pattern.
 - **Average:** Computes the average y-value of the entire pattern and draws a horizontal line at that value.
- **Simulation View:** Set the x-bounds of the overlap simulation plots as one of:
 - **One Swath Width:** Only the mathematically relevant portion of the simulated overlap will be shown, specifically datapoints between $-swathwidth/2$ and $swathwidth/2$.
 - **All Swaths:** All selected simulated adjacent passes will be shown. Can be useful for illustrating the big picture of in-field overlap.

4.5 Report

These settings are persistent but are not saved to the datafile The visual-only settings for the PDF report generated from the **Report** tab. These settings do not affect the content of the report, only the formatting and styling.

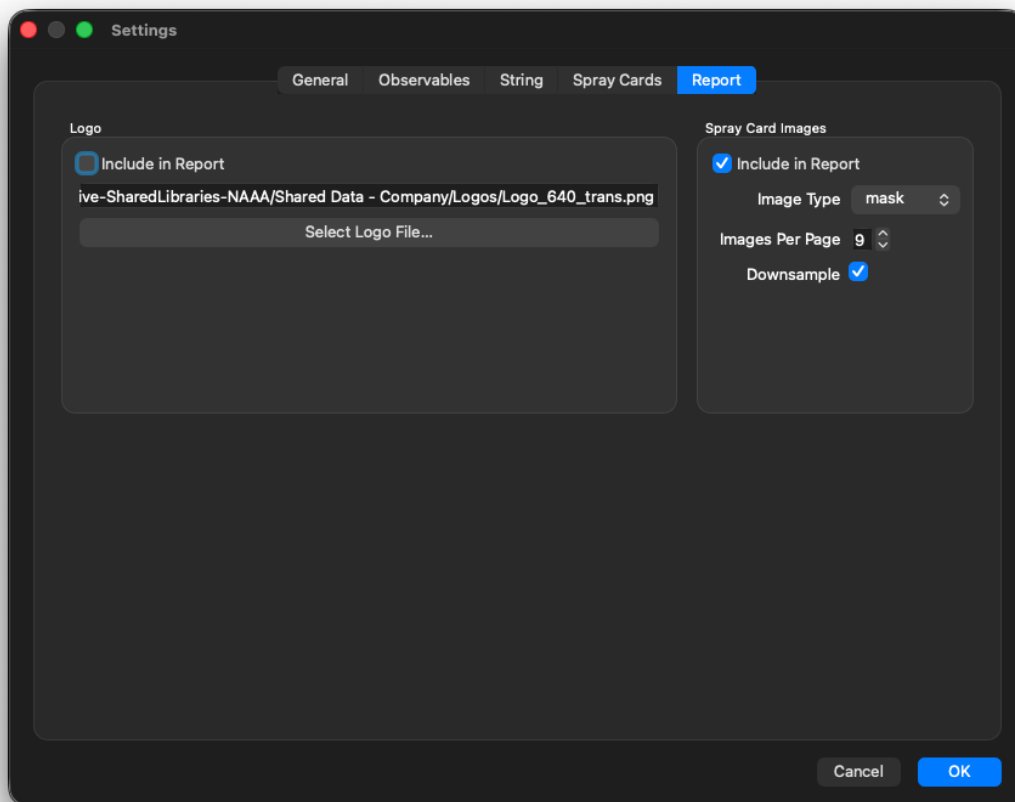


Figure 11: Settings - Report

- **Logo:** A logo image can optionally be included in the report header. Clicking the **Select Logo** button will open a File Chooser to select an image file from anywhere on your computer. The image will be resized to fit a 2-inch wide and 1-inch tall area.
- **Spray Card Images:** The images of the spray cards can optionally be included on dedicated pages in the report. If **Include Spray Card Images** is checked, the user can further configure these as follows:
 - **Image Type:** The type of image to print for each card ([Figure 12](#)):
 - * **Original:** The original scanned image of the card.
 - * **Outline:** The original scanned image of the card with detected stain borders outlined in red.
 - * **Mask:** A representation of the binary mask used to identify stain pixels, where

background pixels are shown in white and stain pixels are shown in blue (if used in coverage AND droplet spectrum analysis), pink (if used in coverage analysis only (such as edge-stains)), or red (if fully excluded from analysis for being under the min stain size).

- **Images per Page:** The number of card images to include on each page of the report.
- **Downsample:** If checked, the card images will be downsampled to 72 dpi to reduce file size. If left unchecked, the original resolution of the card images will be included in the report, which can result in a very large file size.

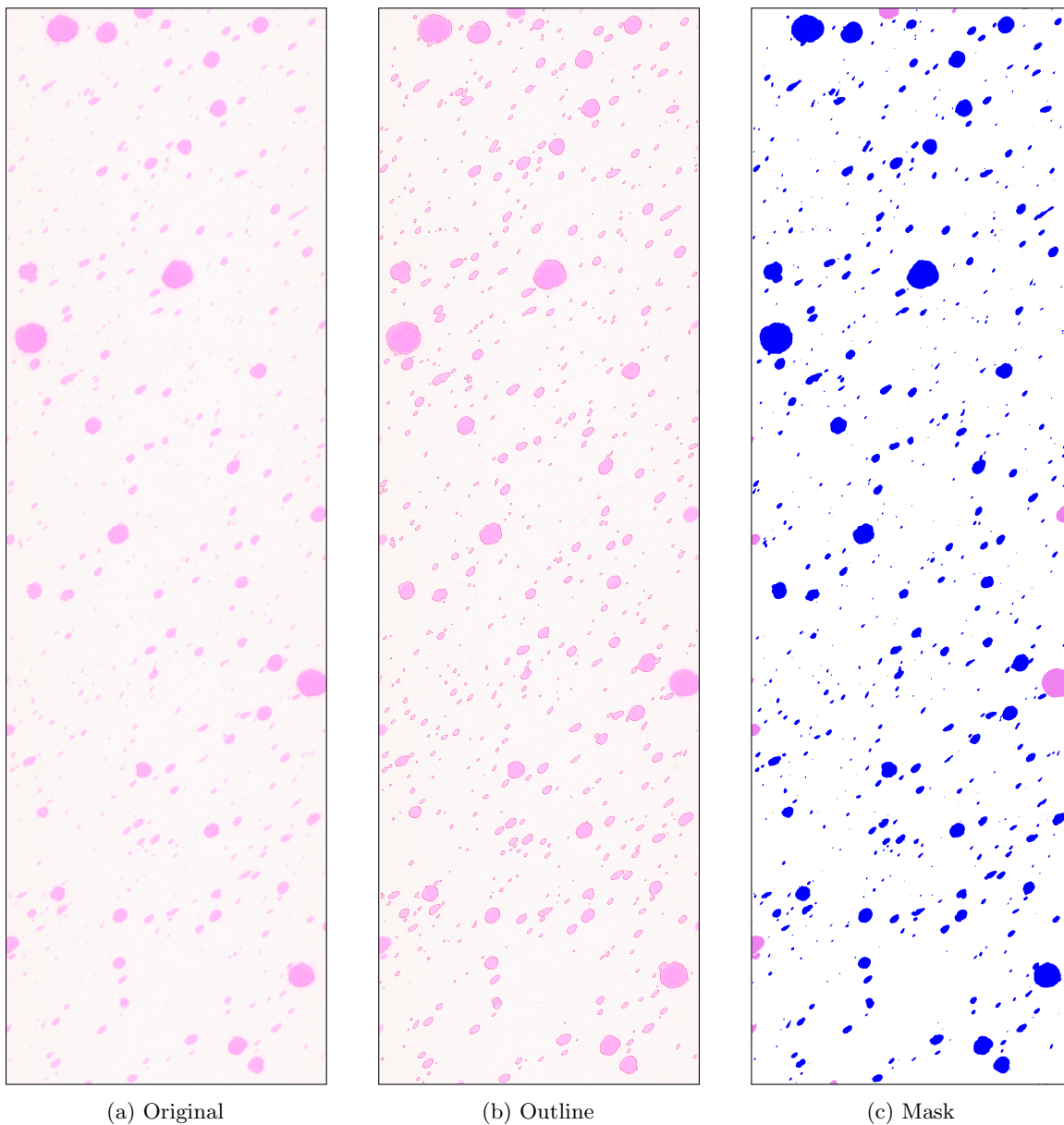


Figure 12: Settings - Report - Spray Card Image Type

5 Creating a New Series

All data and operations in *AccuPatt* require a series. To acquire new data, a new series must be created. From the menu bar, use **File** → **New Series** to view options available (Figure 13). A series is created with the user-defined default number of passes.

- **New Series** → **New Aircraft**: Create an empty series with required parameters set to default values.
- **New Series** → **File Aircraft**: Create a series, copying all application info from a previous data file. Increments the series number by 1. A list of previous data files in your current working directory will appear in the menu for you to select from. Alternatively, choosing **Select File Aircraft** will open a File Chooser allowing you to select a datafile from anywhere.

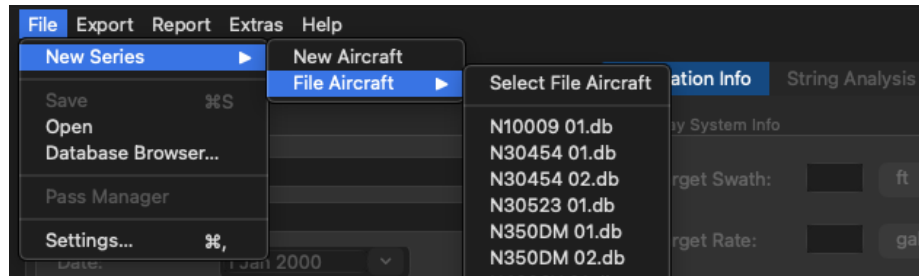


Figure 13: New Series

6 Application Info

It is generally desirable, but not required, to enter in some amount of information about the aircraft, spray system configuration, and/or applicator information to accompany pattern testing data. The **Application Info** tab provides fields to facilitate this (Figure 14). See the [Data Structure](#) section for more details on specific fields.

Here are a few helpful notes regarding the **Application Info** tab:

- The **Fly-In Info** fields are stored persistently.
- There is button to the right of the **Pilot** field which, if hovered over, reads **Load Business Info from File**. Clicking this will prompt you to select a previous datafile, from which all fields within the **Applicator Info** section (excluding Pilot) will be filled.
- When an aircraft make and model are selected, the wingspan field is automatically populated if available.
- The Target Swath units dropdown directs the location (x-axis) units for all string and card plots.
- The **Nozzle Info** section allows adding any number of unique nozzle sets.

The screenshot shows the 'Application Info' tab in the AccuPatt 2.1.1 software. The interface is divided into several sections: Fly-In Info, Applicator Info, Aircraft Info, Spray System Info, Nozzle Info, Pass Observables, and Notes. The Fly-In Info section includes fields for Event Name, Location, Date, and Analyst(s). The Applicator Info section includes fields for Pilot, Business, Street, City, State, ZIP, Phone, and Email. The Aircraft Info section includes fields for Reg. #, Series #, ID, Make, Model, Wingspan, and Winglets. The Spray System Info section includes fields for Target Swath, Target Rate, Boom Pressure, Boom Width, Boom Drop, and Nozzle Spacing. The Nozzle Info section includes fields for Set 1, Type, Size, Nozzle Angle, and Quantity. The Pass Observables section includes a table with columns for Pass 1, Pass 2, and Pass 3, and rows for Ground Speed, Spray Height, Pass Heading, Wind Direction, Wind Speed, Temperature, and Humidity. The Notes section includes fields for Setup Notes and Analyst Notes. The bottom of the window shows the current file path and the last save time.

	Pass 1	Pass 2	Pass 3
Ground Speed	159 mph	171 mph	165 mph
Spray Height	15 ft	13 ft	13 ft
Pass Heading	246°	246°	246°
Wind Direction	214°	243°	257°
Wind Speed	2 mph	3 mph	3 mph
Temperature	60 °F	60 °F	60 °F
Humidity	59 %	59 %	59 %

Figure 14: Application Info Tab

6.1 Pass Observables

In addition to **Application Info**, which is collected on a per-series basis, there is some observable data which is typically collected on a per-pass basis. These are frequently referred to here and, in the program as **Pass Observables** since they consist solely of measured values obtained at the time of the application.

These values can be entered in several locations for the convenience of the analyst. They are typically entered in the applicable area of the **Application Info** tab or alternately from either the **Capture/Edit String** window or the **Card Manager** window for any individual pass. Choosing **File → Pass Manager** from the menu will open the **Pass Manager** window (Figure 15) which contains a table with rows for each pass and columns for each pass observable parameter. In addition to the editable fields for pass observables, the pass name itself may be changed here. There are also checkboxes to facilitate the inclusion/exclusion of string and/or card data from series-wise calculations and plots.

The **Pass Manager** window also provides buttons to add or remove passes from the series. There is no functional limit to the number of passes per series.

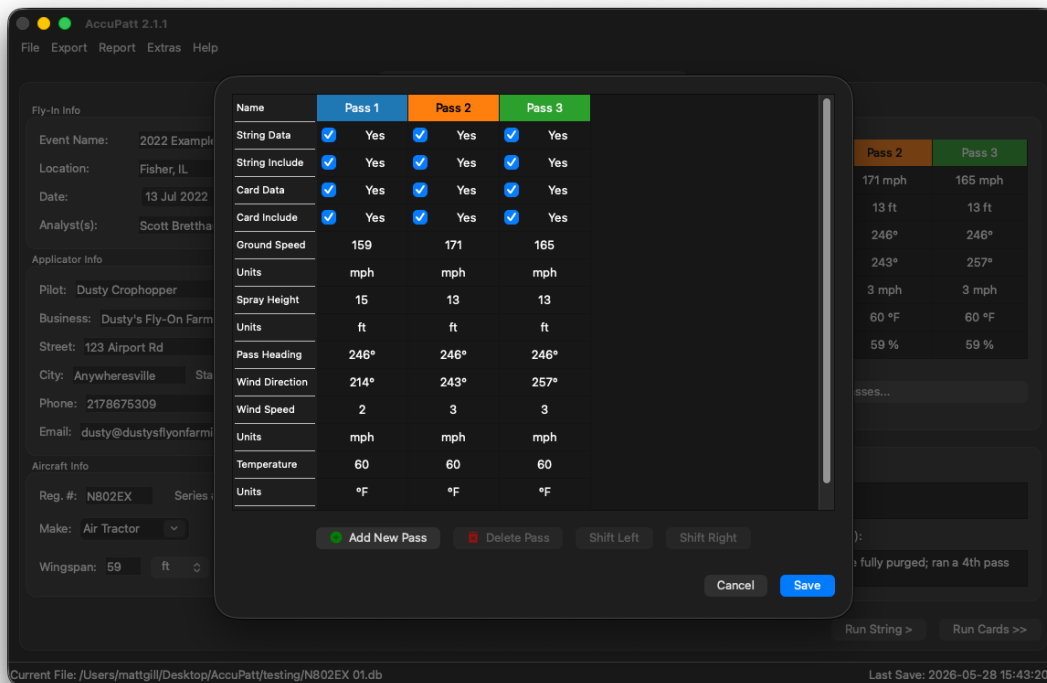


Figure 15: Pass Manager

7 String Analysis

String data is collected on a per-pass basis but may be analyzed on a per-series basis. The **String Analysis** tab provides a list of passes in the series and pass/series processing options. This tab also contains sub-tabs for viewing/modifying data as discussed below.

7.1 Capture Pass

Choose the pass to be collected from the list of passes, then click the **Capture Pass** button. This will open the **Capture/Edit Pass** window (Figure 16).

Indicators for the connection status of the string drive motors and spectrometer are provided along with the selected Dye and calculated spatial resolution (based on string speed and integration time). A shortcut to **String Settings** is also provided for convenience (See [section 4.3](#)).

If both the string drive motors and spectrometer are connected, clicking the **Start** button will engage the string drive, advancing the string, and simultaneously will change its text to **Mark**. Clicking the **Mark** button will begin the collection of spectrometer data at intervals specified by the set **integration time**. Data will continue to be collected until the set **string length** is reached. At any time after data collection is begun, clicking the **Abort** button will stop the string drive and clear all data collected up to that point.

The **Clear** button may be used to clear the pass collected string data whenever data collection is not in progress. If string data exists for the current pass, it will ask for confirmation first.

Pass Observables can be entered at the right at any time if not already recorded in the **Application Info** tab.

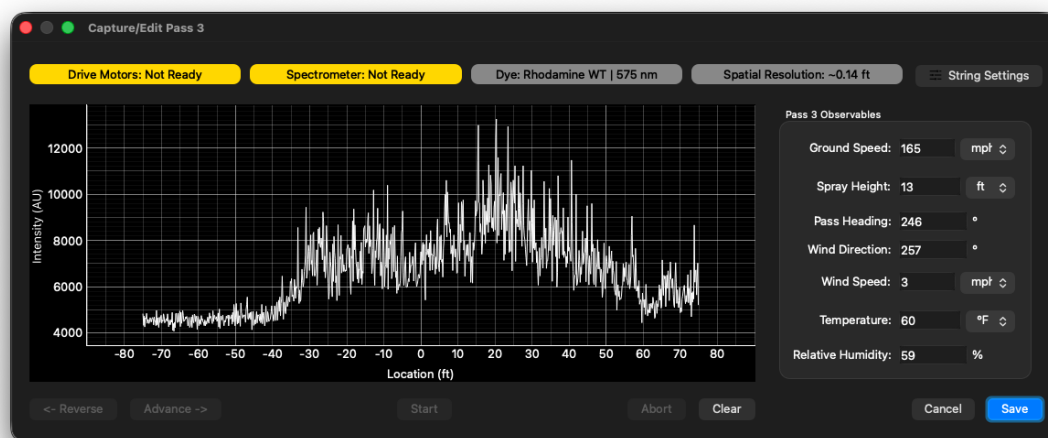


Figure 16: Capture/Edit Pass Window

7.2 Individual Passes Tab & Pass Processing

Once string data for a pass is collected, the pass will have a check next to it in the pass list on the **String Analysis** tab. When a pass with string data is selected from the pass list, the **Individual Passes** sub-tab (Figure 17) will show two plots discussed below.

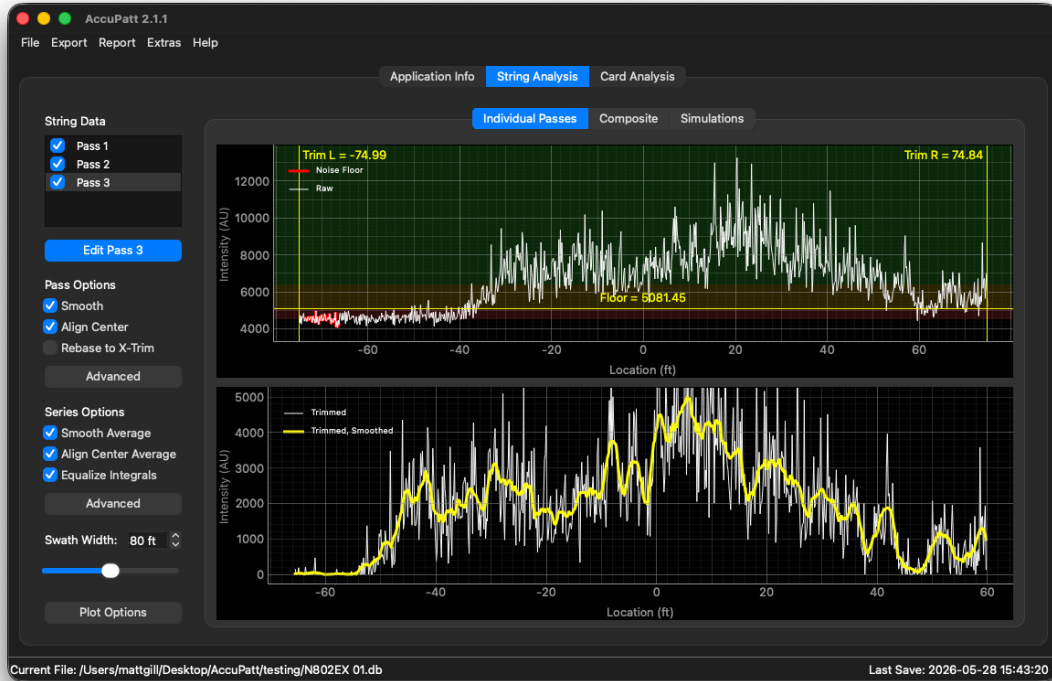


Figure 17: String Analysis - Individual Passes

The **Interactive Trim Plot** [Upper Plot] shows the data exactly as acquired. There are two vertical yellow lines (**Trim L** and **Trim R**) which can be dragged horizontally into the pattern to set the left and right boundaries, respectively, of data to be eligible for analysis. This is useful to exclude edge spikes/dips from artifacts such as green/red marks from data collection. The yellow horizontal **Floor** line can be dragged vertically to set the lower bound for data to be eligible for analysis. This is useful to exclude background noise (i.e. blank string variation) from the pattern.

By default, the **Floor** is algorithmically set to remove background noise from blank string using a Signal-to-Noise Ratio (SNR) where data points with $SNR < 3$ are excluded ([section 7.2.1](#)).

As an additional assistance to the analyst in determining the strength of a pattern, background highlighting is provided based on SNR. Where $SNR < 3$, the background is red to represent great uncertainty of whether data points are signal or noise. Where $3 \leq SNR < 10$, the background is yellow to represent mild uncertainty of whether data points are signal or noise. Where $SNR \geq 10$, the background is green to represent certainty that data points are signal and not noise.

The **Processed Preview Plot** [Lower Plot] shows the result of applying trim/floor from the **Interactive Trim Plot** to the raw data. Additionally, it shows the smoothed version of this data in yellow. The only difference between the smoothed version here and what is shown in subsequent

plots is the application of horizontal centering. See [section 7.5](#) for details on individual pass settings.

7.2.1 Automated Pattern Isolation & Denoising

After trimming edge anomalies, it is necessary to raise the **Floor** of the pattern to exclude data subjectively perceived to be blank string. This step is critical to ensuring that this background noise is not included in the patterns used for interpretation and uniformity calculations. This is really no different to adjusting physical filters on the fluorometers of the bad-old days, except that this step is now software post-processing instead of hardware pre-processing.

To counter the subjective nature of this step and combat its inadvertent omission by less-experienced users, *AccuPatt* automatically sets the initial **Floor** to a Signal-to-Noise Ratio (SNR) of 3. Because every pattern is different and the algorithm isn't perfect, the **Floor** may still be adjusted freely after it is initially set.

7.2.1.1 Identifying a Segment of Blank String The lowest contiguous 5% of raw pattern data (after excluding data outside the **Trim L** and **Trim R** bounds) is found and used as a representation of blank string. This is graphically represented to the user by highlighting that segment of data points in red. *Note: This depends upon and assumes the string actually having a segment of string without dye deposit; if this is not the case, do not rely upon this algorithm.*

7.2.1.2 Compute the Baseline The baseline for SNR is the y-value to which all data points will be compared to.

$$SNR_{base} = \bar{y} \quad (1)$$

where:

$\bar{y}$ is the mean value of the data points in the blank string segment.

7.2.1.3 Compute the Noise of Blank String We can determine the noise as the standard deviation of the blank string data by:

$$N_{RMS} = \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n - 1}} \quad (2)$$

where:

y_i are the individual data points in the blank string segment, and
 n is the number of data points in the blank string segment.

7.2.1.4 Set the Floor to a SNR=3 The SNR is computed as:

$$SNR = \frac{Y - SNR_{base}}{N_{RMS}} \quad (3)$$

where:

Y is any given y-value.

So, using an SNR=3, the floor is computed as:

$$Floor = 3N_{RMS} + SNR_{base} \quad (4)$$

7.2.2 Assessing Quality of Pattern Data

Before a pattern is interpreted, its data should be assessed for quality. Specifically, the goal should be to have reasonable certainty that the deposited pattern has been isolated from the noise of blank string. Below are some examples of what to look for. The main takeaway is to question the validity of the data before you question the shape/width of the resultant pattern.

7.2.2.1 Good Isolation A pattern with good isolation from blank string is easy to spot (Figure 18). The segments of blank string are visible as fairly flat compared to the pattern with the vertical scale of the plot squeezing the vertical variation in the blank string segments to appear as barely perceptible. Note also that a majority of the pattern is in the green zone ($SNR \geq 10$), meaning that the pattern signal is very clearly differentiated from the variation in blank string noise.

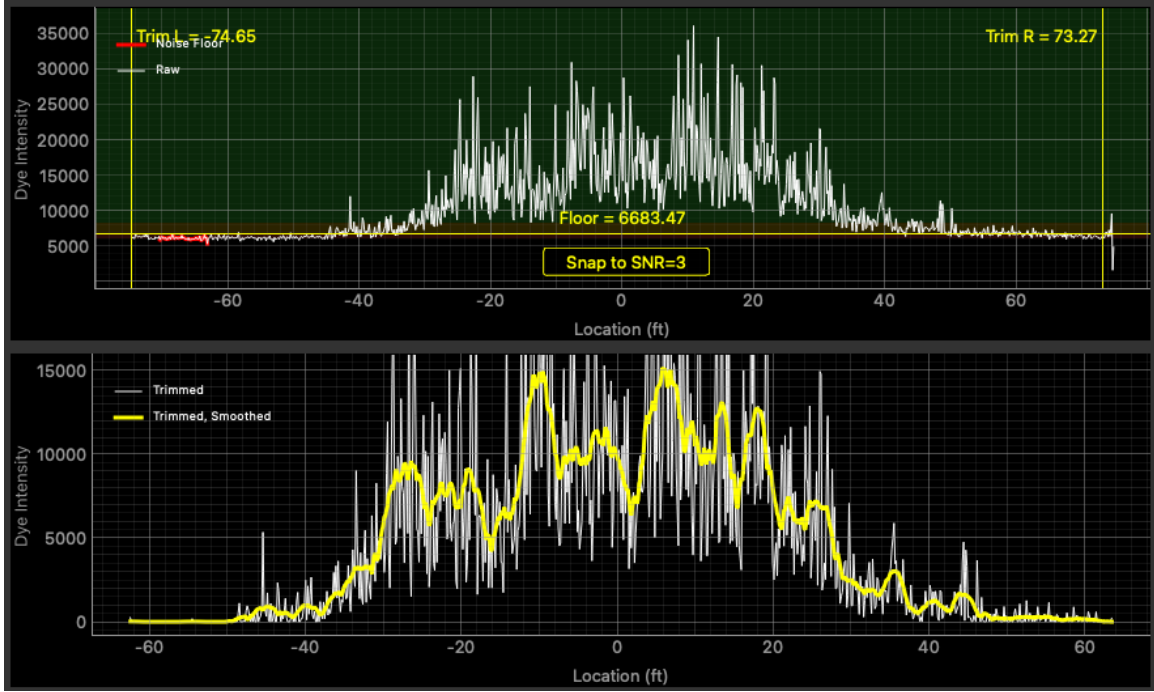


Figure 18: String Analysis - Quality of Pattern Data - Good Isolation

7.2.2.2 Marginal Isolation A pattern with marginal isolation from blank string can be difficult to spot (Figure 19). The segments of blank string are somewhat visible, but are not quite visibly differentiable from the pattern; in other words it is difficult to visibly determine the point at which the pattern begins on at least one side (left or right). The vertical variation in what *AccuPatt* has determined to be the blank segment is very perceptible. Note also that a majority of the pattern is in the yellow zone ($3 \leq SNR < 10$), meaning that there is mild uncertainty that the pattern is isolated from the blank string segment.

The reasons for marginal isolation are myriad and include, but are not limited to:

- Insufficient dye loading
- Degraded or improperly mixed dye
- Testing exceedingly large droplet spectra, such as AccuFlo nozzles where the string collection efficiency is low.
- Insufficient sampling width; if dye actually deposits onto the entire extent of the string, there is no blank string to reference from.

In the case of the provided example in Figure 19, it was determined that there actually was deposition across the entire extent of the string. This determination was made based on an observed $SNR_{base} \approx 7400$ for other passes in the same series. Thus, in this case, the noise isolation algorithm's suggested

Floor was not appropriate and could have been overrode. However, because the pattern could not be isolated it was excluded from analysis.

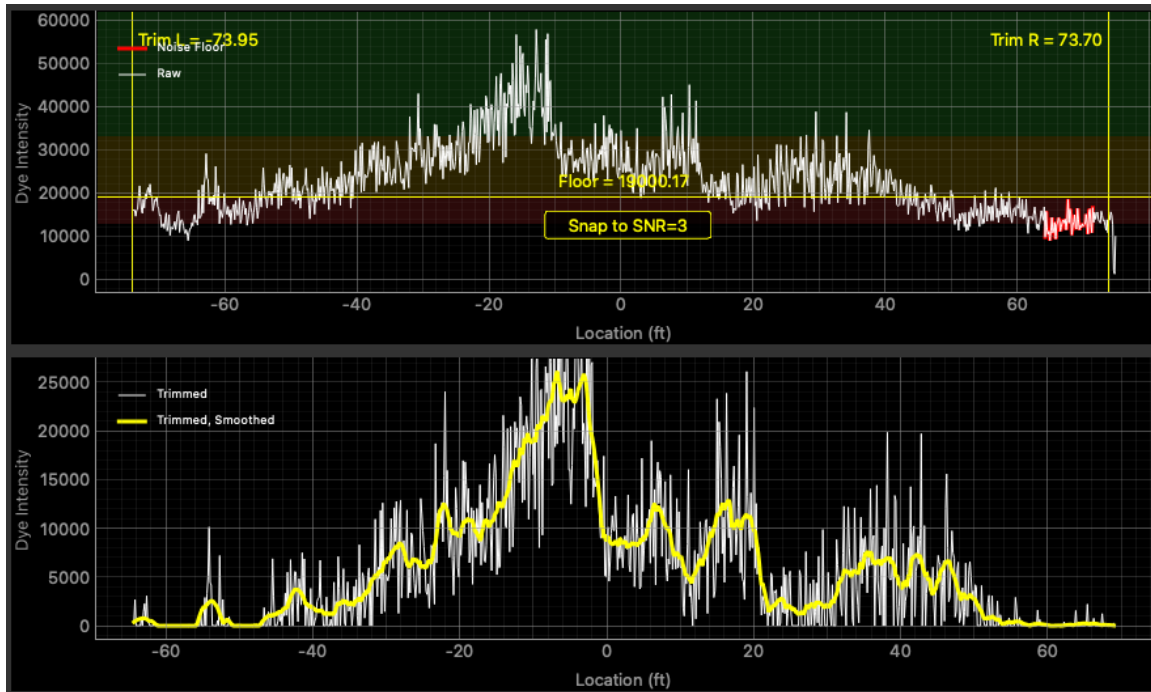


Figure 19: String Analysis - Quality of Pattern Data - Marginal Isolation

7.2.2.3 Insufficient Isolation A pattern with insufficient isolation will have little to no visually distinguishable transition point between itself and blank string (Figure 20). The majority of the string's extent will have similar vertical variation and will appear exaggerated due to the scaling of the plot in absence of a pattern rising out from it. Note that a majority of the data points do not even survive the $SNR \geq 3$ Floor.

It may seem obvious, but these "patterns" can and have been interpreted as actual deposition by experienced folks who simply failed to question the data.

The reasons for insufficient resolution are typically:

- No dye loading
- Incorrect or severely degraded dye
- Incorrect excitation light source and filter
- Incorrect target emission wavelength or integration time set (see [section 4.3.1](#))

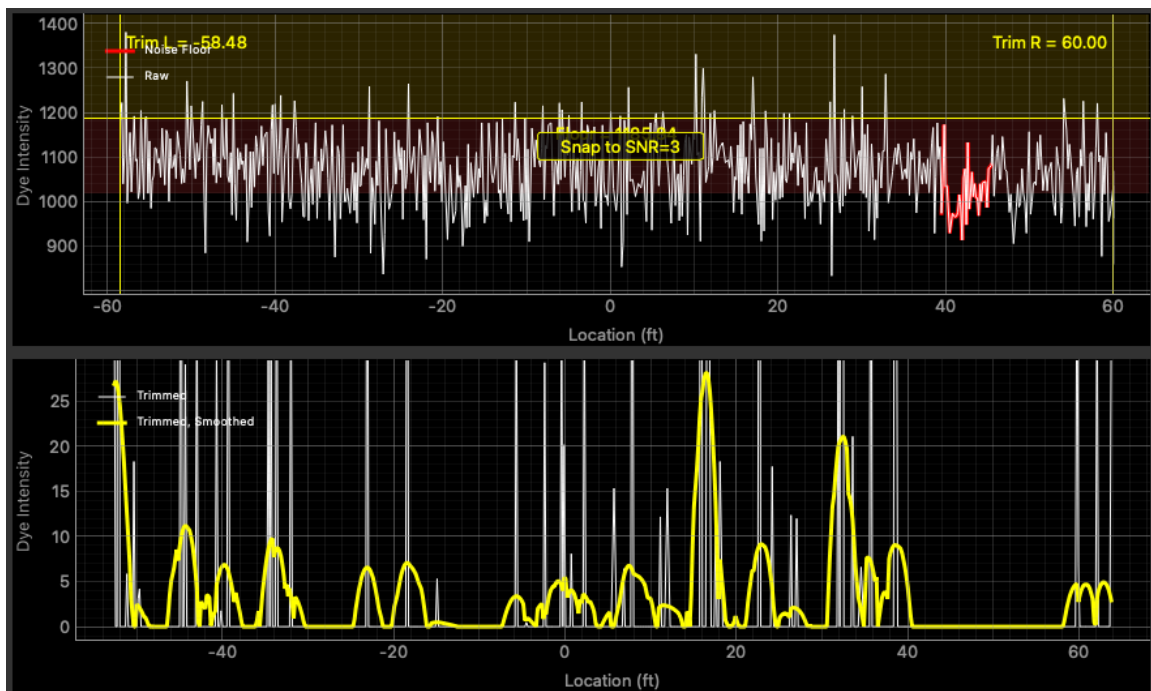


Figure 20: String Analysis - Quality of Pattern Data - No Isolation

7.3 Composite

The **Composite** sub-tab contains two plots (Figure 21):

- **Overlay:** [Upper Plot] Each individual (non-excluded) pass is plotted together.
- **Average:** [Lower Plot] Each individual (non-excluded) pass is used to generate an average pattern. A dashed overlay can be optionally included based on the **Average Overlay** setting (See [section 4.3.3](#)).

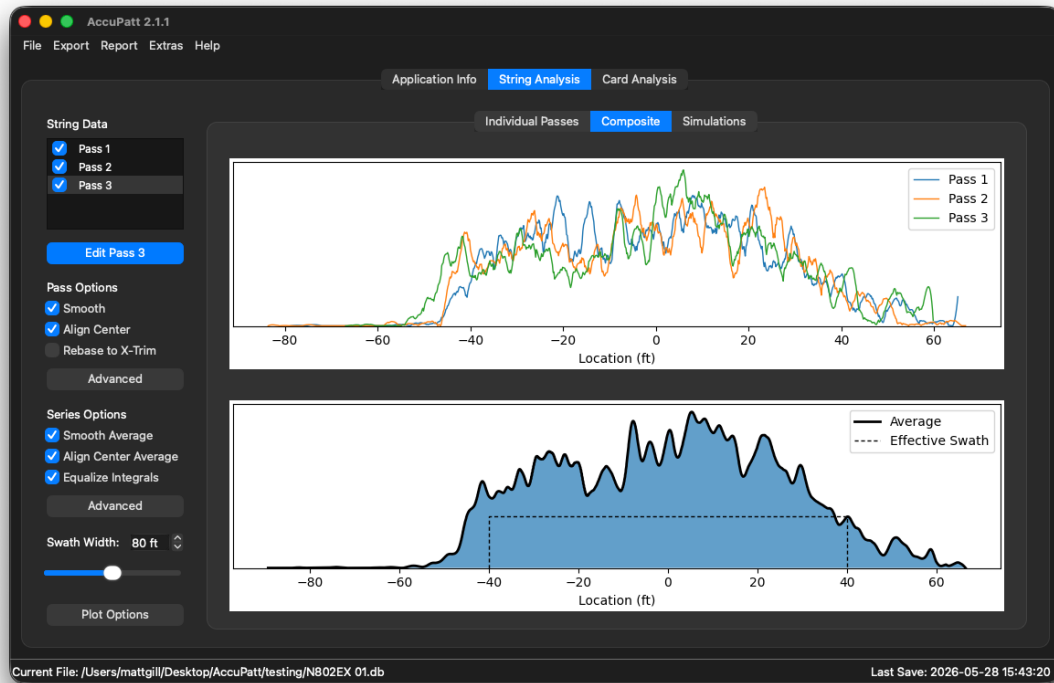


Figure 21: String Analysis - Composite

7.4 Simulations

The **Simulations** sub-tab contains two plots (Figure 22). Both plots simulate additively overlapping the measured pattern at intervals of the adjusted swath width. The bold black line represents the resultant summed pattern, and is interpreted here as projected in-field uniformity. The dashed horizontal line represents the mean value, about which standard deviations are computed for coefficient of variation (CV) calculations. In other words, the further a data point on the bold black line is (vertically) from the dashed line, the more it contributes to the CV.

- **Racetrack:** [Upper Plot] Each simulated pass is plotted going the same direction (into the screen/paper)
- **Back & Forth:** [Lower Plot] Every odd simulated pass is mirrored about the center. **The back & forth simulation will not accurately represent in-field uniformity if the base pattern is obtained in a crosswind.**

The number of simulated adjacent passes is determined automatically based on the adjusted swath width and pattern width. The plots will denote the number used when it exceeds the legend's available area.

The visible x-value range of the plot windows can be set to show only the mathematically relevant range (single swath width) or all simulated passes (see section 4.3.3).

A table is included which calculates the CV for select swath widths near the set adjusted swath width.

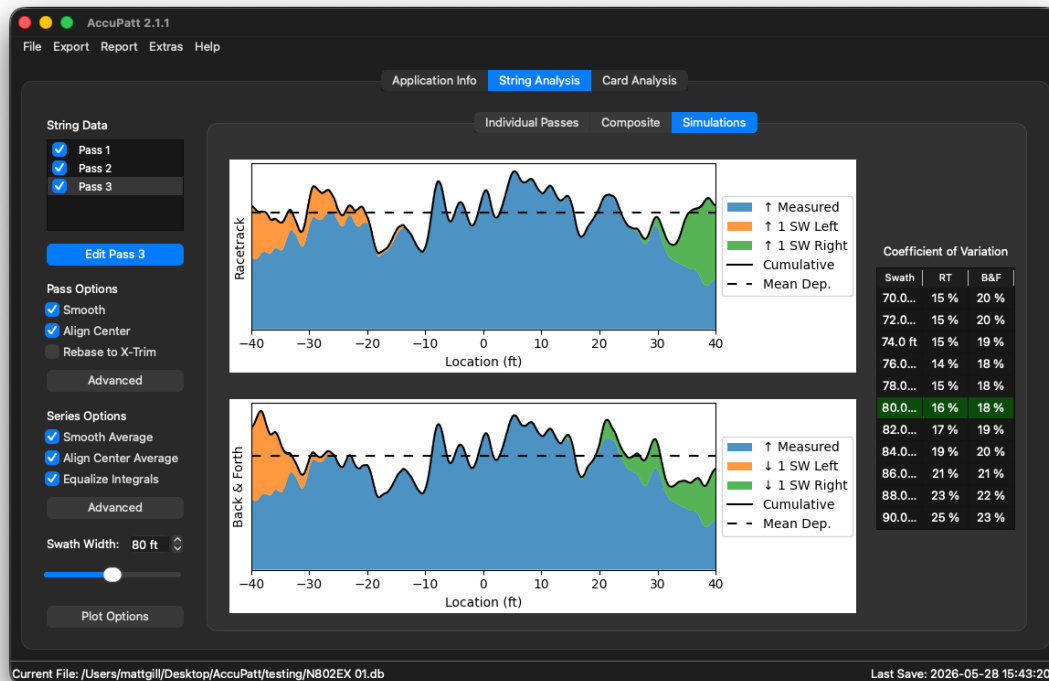


Figure 22: String Analysis - Simulations

7.5 Pattern Processing Options

Pass Options affect only the selected pass:

- **Include In Composite:** Once a pass is recorded, a checkmark will appear adjacent to it in the pass list. This indicates that it is recorded and included in composite (series-wise) calculations and plots. Clicking the checkmark will turn it into a dash, indicating that it is recorded, but excluded from composite calculations and plots.
- **Smooth:** Apply smoothing filter to pass. Checked by default.
- **Align Center:** Shift pattern left/right using chosen center method. Checked by default.
- **Rebase to X-Trim:** Recalculate each data point location such that the trimmed length is equal to the captured string length. If string speed is slightly faster than originally calibrated, this facilitates post-process correction to ensure accurate swath measurement. Unchecked by default.
- **Advanced Options:** Opens a popup with the following options (defaults as set in Settings/String ([section 4.3.2](#))):
 - **Smoothing Window:** X-domain based distance over which the sliding smoothing filter is fit and applied
 - **Smoothing Order:** Polynomial order for smoothing filter
 - **Center Method:** Choose whether to center the pattern with the centroid or Center of Distribution.

Series Options can apply to all passes and/or the average pattern:

- **Smooth Average:** Apply smoothing filter to average pattern. Checked by default.
- **Align Center Average:** Shift average pattern horizontally using chosen center method. Checked by default.
- **Equalize Integrals:** Scale each pass to the same integrated area. Checked by default.
- **Advanced Options:** Opens a popup with the following options (defaults as set in Settings/String ([section 4.3.2](#))):
 - **Smoothing Window:** X-domain based distance over which the sliding smoothing filter is fit and applied
 - **Smoothing Order:** Polynomial order for smoothing filter
 - **Center Method:** Choose whether to center the pattern with the centroid or Center of Distribution.

Swath Width defaults to the target swath width but is saved independently as the **Adjusted Swath Width**. This is the value that drives the average overlay (if inner-swath-half-average is selected), simulations and CV calculations.

The **Plot Options** button opens a popup with alternative, but equivalent, control to those in [section 4.3.3](#). These options have no mathematical effect on the analysis and are merely ways to adjust the generated graphics.

8 Spray-Card Analysis

Card data is collected on a per-pass basis but may be analyzed on a per-card or per-series basis as well. The **Card Analysis** tab provides a list of passes in the series. This tab also contains sub-tabs for viewing/modifying data as discussed below.

8.1 Capture Pass

In the **Card Analysis** tab, click the **Capture/Edit Pass** button to open the **Card Manager** window (Figure 23). Pass Observables can be entered at the right at any time if not already recorded in the **Application Info** tab.

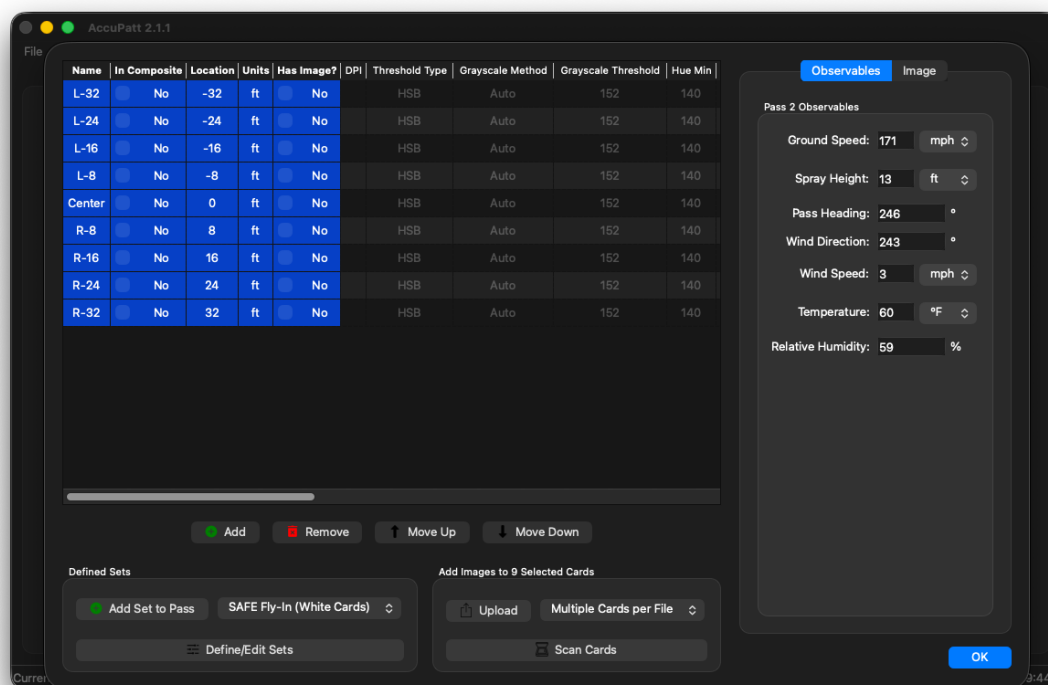


Figure 23: Spray-Card Analysis - Card Manager

8.1.1 Adding Cards to Pass

The first step is to declare/add the cards to the Pass. They can be declared/added one-by-one or via **Defined Sets**. This includes setting their names and locations (if desired). Cards are represented by rows in the table with editable property columns. The rows/cards can be selected individually, as a group by click-dragging over multiple rows or as a group by holding Shift (continuous selection) or Ctrl/Cmd (non-continuous selection). There are four buttons below the table:

- **Add:** Adds a new card in the last row, setup with default Thresholding (section 4.4.2), Stain Processing (section 4.4.3) and Spread Factor (section 4.4.4) settings.
- **Remove:** Removes all selected cards. If any selected cards have image data in the datafile, a confirmation will appear prior to removal.

- **Shift Up:** Shift selected cards up by one row, if possible
- **Shift Down:** Shift selected cards down by one row, if possible.

Below these buttons are controls for utilizing **Defined Sets** which facilitate adding a pre-configured list of spray-cards to the list. Use the dropdown to select a set and click **Add Set To Pass** to append the cards to the table. More details can be found in [section 4.4.5](#).

8.1.2 Uploading Images to Cards

The next step is to select the cards you want to upload images to (see above selection procedure). At the bottom of the window are controls to add images to the selected cards in the table. A dropdown facilitates choice of the upload method (see [section 4.4.1](#)) and clicking the **Upload** button will open a file chooser window to select all applicable files. *Note: Image files must be of type *.png, *.tif or *.tiff. Ensure that images are acquired with no pre-processing, corrections, enhancements or compression.*

8.1.2.1 One Card Per File For images which are already cropped appropriately, they can be directly uploaded to the datafile ([Figure 24](#)). Selected files [right column] can be dragged to alter the order to match the proper order of declared card names. An **Assert DPI** menu allows setting of image DPI as automatic (set by image metadata) or to one of several common values. It may also optionally be supplied via direct-type entry to be applied to all uploaded cards.

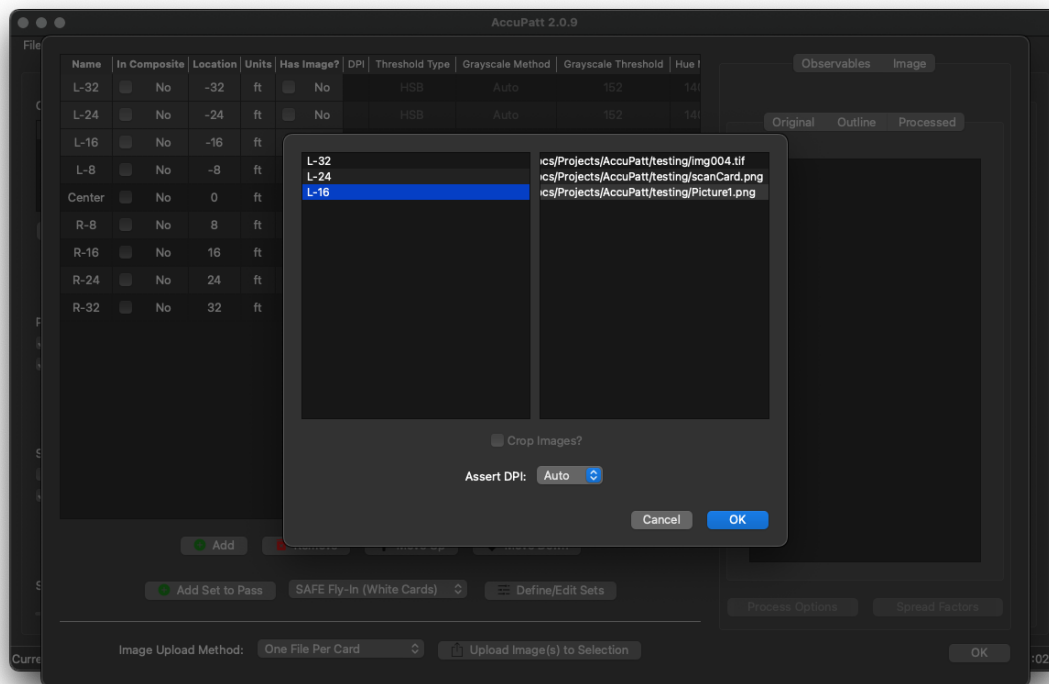


Figure 24: Spray-Card Analysis - Upload Single Images

8.1.2.2 Multiple Cards Per File When multiple cards are acquired in a single large scan, this method will allow cropping of each card and saving individually to the datafile. After selecting the file, the image will be shown with each card's independently draggable, resizable region-of-interest (ROI) (Figure 25). The image may be panned or zoomed-upon. To decrease RAM consumption, large images are shown in lower resolution here, but full-resolution ROIs will be cropped out of the original image using a scaled coordinate system.

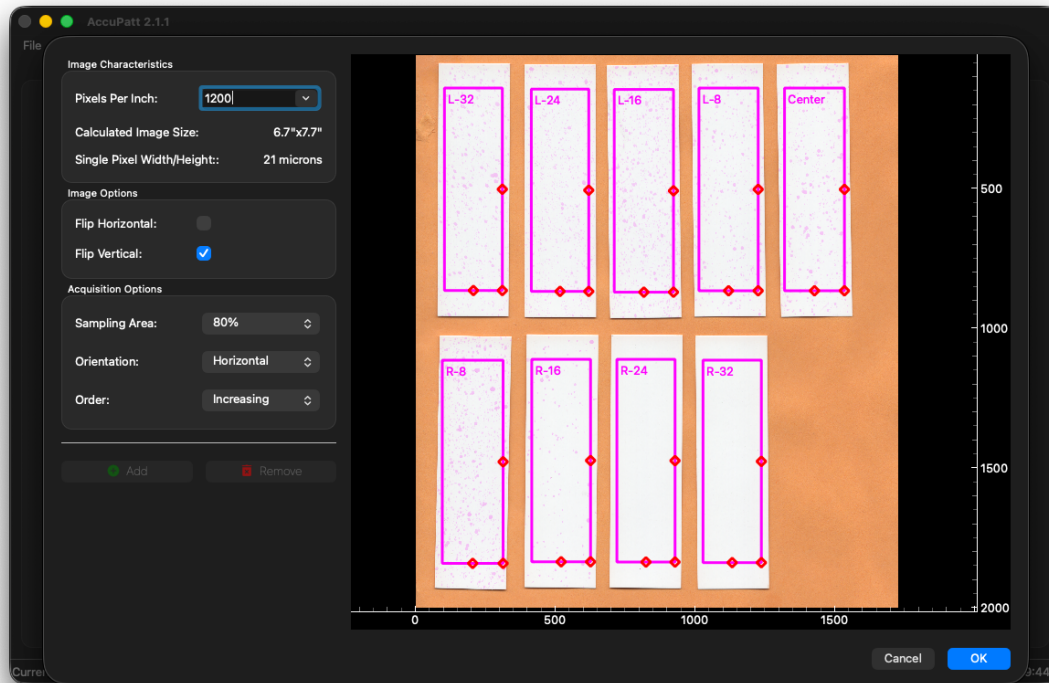


Figure 25: Spray-Card Analysis - Upload Image with Multiple Cards

The **Pixels Per Inch** is automatically set by the image metadata, but user-adjustable. This is used in conjunction with the image pixel dimensions to show calculated image dimensions. These are provided only to allow verification by the user of proper pixels per inch setting.

The image acquisition options adjust the method by which the ROIs are drawn. They are persistent settings discussed in [section 4.4.1](#).

In the case that ROIs are not automatically drawn properly, the **Add** and **Remove** buttons may be used as follows:

- **Add:** Click the **Add** button. The cursor will change to a crosshair. Press the left mouse button on one corner of the spray-card, drag the cursor to the other corner, then release the left mouse button. A new ROI will be drawn (scaled per the **Sampling Area** set above) and all ROIs will be relabeled as necessary (per the **Orientation** and **Order** set above). Note that if your original selection of spray-cards (selected from the Card Manager) already each have ROIs drawn, no more may be drawn.
- **Remove:** First, click an ROI you would like to remove. The **Remove** button will become enabled, and its text will change to "Remove [spray-card.name]." Clicking the button will

remove the ROI and all ROIs will be relabeled as necessary (per the **Orientation** and **Order** set above).

You can manipulate each ROI manually to fine-tune what part of the card to sample:

- **Drag:** Click and hold within the ROI to reposition it.
- **Adjust Width:** Use the red handle on the right edge of the ROI to adjust only the width.
- **Adjust Height:** Use the red handle on the lower edge of the ROI to adjust only the height.
- **Adjust Height/Width Together:** Use the red handle on the lower right corner of the ROI.

8.1.2.3 From ZIP This option is primarily present to ease importing Swath Gobbler™ data which saves sequentially identified images in a ZIP archive. After selecting the ZIP file, images are unpacked and fed to the **One Card per File** interface (see [paragraph 8.1.2.1](#)).

8.1.3 Verify Image Processing Efficacy

Once a card has image data associated with it, selecting its row in the table will make a preview image visible in the **Image** tab at the right of the window (Figure 26). The image may be viewed as follows (see Figure 12 for a visual representation):

- **Original:** The unmodified original image.
- **Outline:** The original image forms the base layer. All stains have their perimeter outlined.
- **Processed:** A blank image with stains drawn (filled) and colored according to the following:
 - **Undersize:** [Red] Not counted toward coverage or droplet spectrum analysis.
 - **Edge:** [Pink] Counted toward coverage, but not droplet spectrum analysis.
 - **Valid:** [Blue] Counted toward coverage and droplet spectrum analysis.

The purpose of these image views is to subjectively verify the image processing efficacy. Adjusting the **Threshold** and **Stain Processing** settings for the card in the table will trigger the preview image to reprocess. A more interactive adjustment is described below.

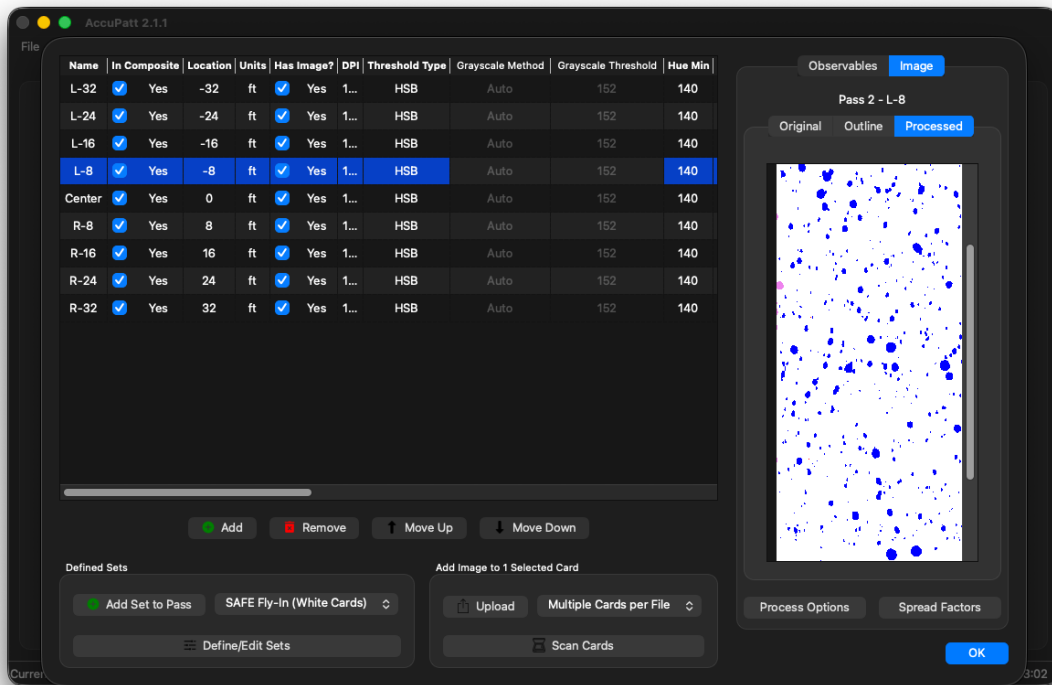


Figure 26: Spray-Card Analysis - Card Manager

8.1.4 Threshold and Stain Processing Adjustments

If desired, a selected card's **Threshold** and **Stain Processing** settings may be adjusted by clicking the **Process Options** button below the card preview image which opens an interactive window with both the **Outline** and **Processed** images side-by-side for subjective review (Figure 27).

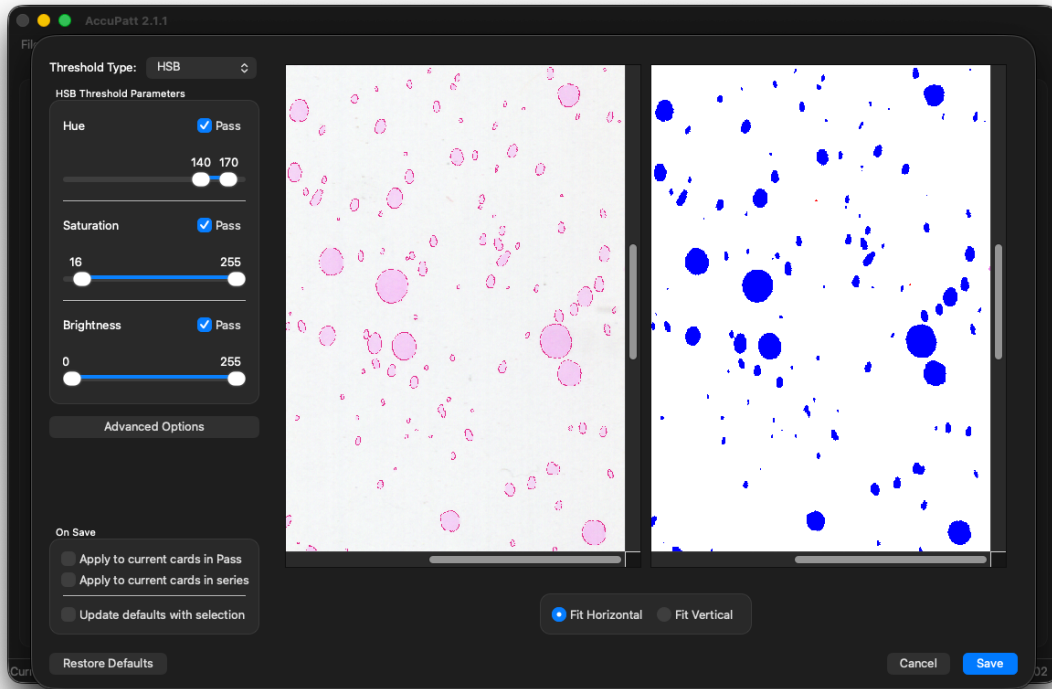


Figure 27: Spray-Card Analysis - Card Manager - Process Options

Discussion of the **Threshold** parameters are in [section 4.4.2](#).

Discussion of the **Stain Processing** parameters are in [section 4.4.3](#).

Adjustments are saved only to the selected card unless **Apply to Current Cards in Pass** or **Apply to Current Cards in Series** is checked before saving. Additionally, checking **Update defaults with selection** will change your persistent settings for future cards ([section 4.4.2](#), [section 4.4.3](#)).

8.1.5 Spread Factor Adjustments

If desired, a selected card's **Spread Factor** settings may be adjusted by clicking the **Spread Factors** button below the card preview image which opens an interactive window with a preview of the equation used to compute droplet size from stain size (Figure 28).

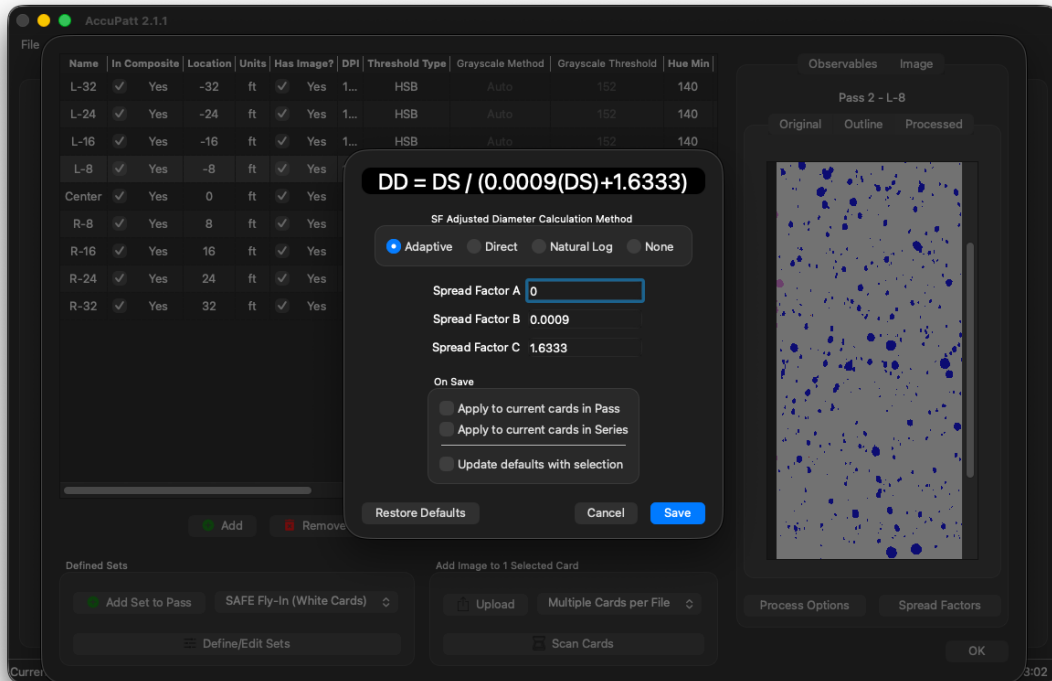


Figure 28: Spray-Card Analysis - Card Manager - Spread Factors

Discussion of the **Spread Factor** parameters are in [section 4.4.4](#).

Adjustments are saved only to the selected card unless **Apply to Current Cards in Pass** or **Apply to Current Cards in Series** is checked before saving. Additionally, checking **Update defaults with selection** will change your persistent settings for future cards ([section 4.4.4](#)).

8.2 Individual Passes Tab

When a pass is selected from the pass list on the **Card Analysis** tab, the **Individual Passes** sub-tab contains a location-domained plot and a table of spray-card statistics (Figure 29). The table's first several columns are editable, so that card name, location, location units and inclusion in composite flag may be set here. The plot will automatically update according to modifications made in the table.

The y-scale of the plot as well as the inclusion of shading are based on the respective options in the **Card Plot Options** window and is discussed in [section 4.4.6](#) and [section 4.4.7](#).

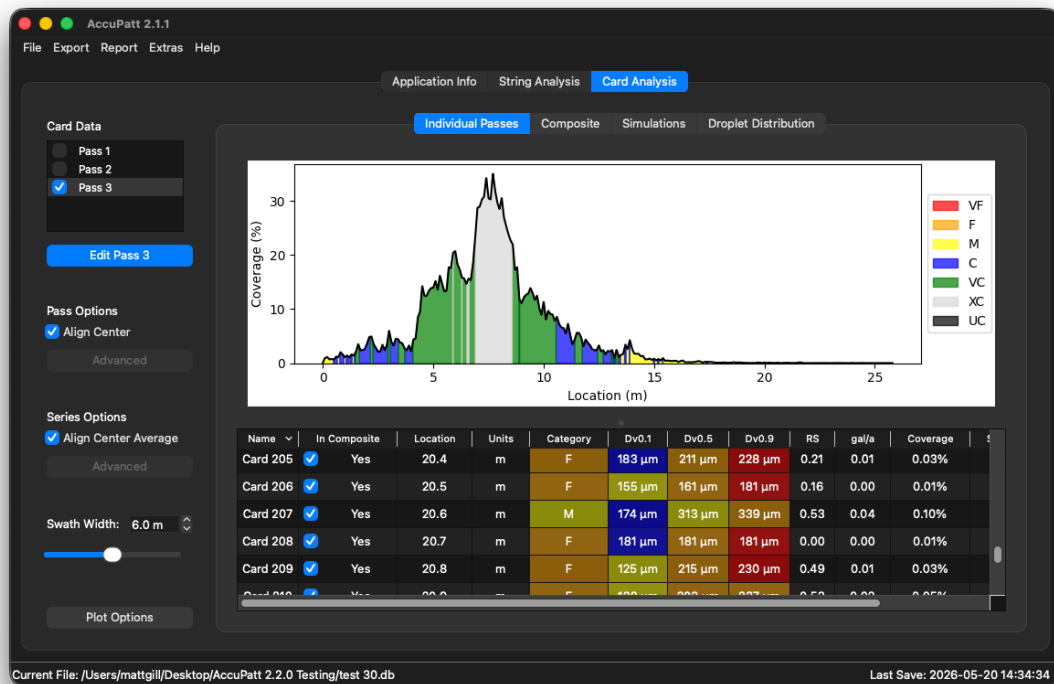


Figure 29: Spray-Card Analysis - Individual Pass Tab

8.3 Composite Tab

The **Composite** tab contains two plots (Figure 30):

- **Overlay:** [Upper Plot] Each individual (non-excluded) pass is plotted together.
- **Average:** [Lower Plot] Each individual (non-excluded) pass is used to generate an average pattern. A dashed overlay and/or shading can be optionally applied based on the respective option in the Card **Plot Options** window and is discussed in [section 4.4.7](#).

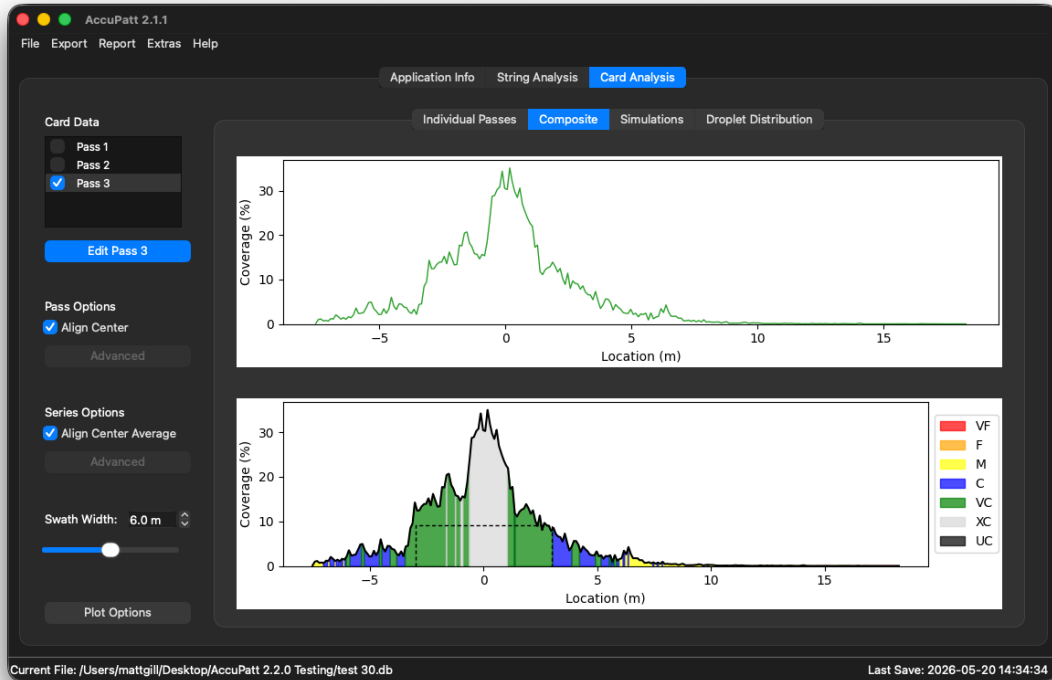


Figure 30: Spray-Card Analysis - Composite View

8.4 Simulations Tab

The **Simulations** sub-tab contains two plots (Figure 31). Both plots simulate additively overlapping the measured pattern at intervals of the adjusted swath width. The bold black line represents the resultant summed pattern, and is interpreted here as projected in-field uniformity. The dashed horizontal line represents the mean value, about which standard deviations are computed for coefficient of variation (CV) calculations. In other words, the further a data point on the bold black line is (vertically) from the dashed line, the more it contributes to the CV.

- **Racetrack:** [Upper Plot] Each simulated pass is plotted going the same direction (into the screen/paper)
- **Back & Forth:** [Lower Plot] Every odd simulated pass is mirrored about the center. **The back & forth simulation will not accurately represent in-field uniformity if the base pattern is obtained in a crosswind.**

The number of simulated adjacent passes is determined automatically based on the adjusted swath width and pattern width. The plots will denote the number used when it exceeds the legend's available area.

The visible x-value range of the plot windows can be set to show only the mathematically relevant range (single swath width) or all simulated passes (see section 4.4.7).

A table is included which calculates the CV for select swath widths near the set adjusted swath width.

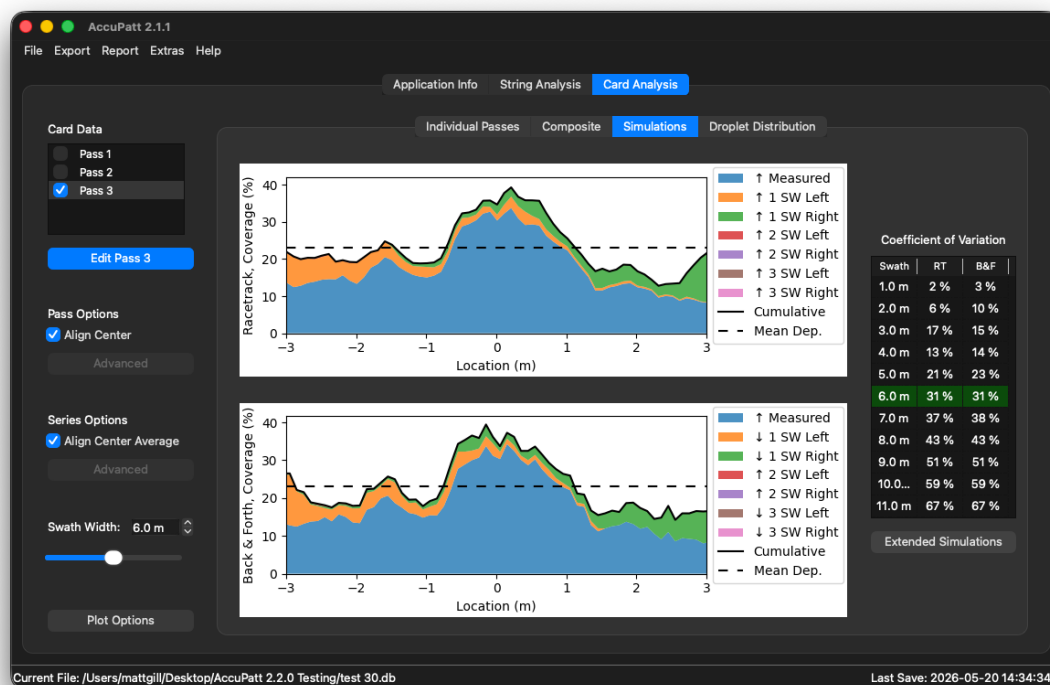


Figure 31: Spray-Card Analysis - Simulations

8.4.1 Extended Simulations

Clicking the **Extended Simulations** button will open a window wherein a range of swath widths can be simulated and plotted to show both their CV and in-swath min/mean/max coverage (Figure 32). Setting targets of Min/Max coverage and Max CV allows the plot to highlight swath widths which meet those criteria.

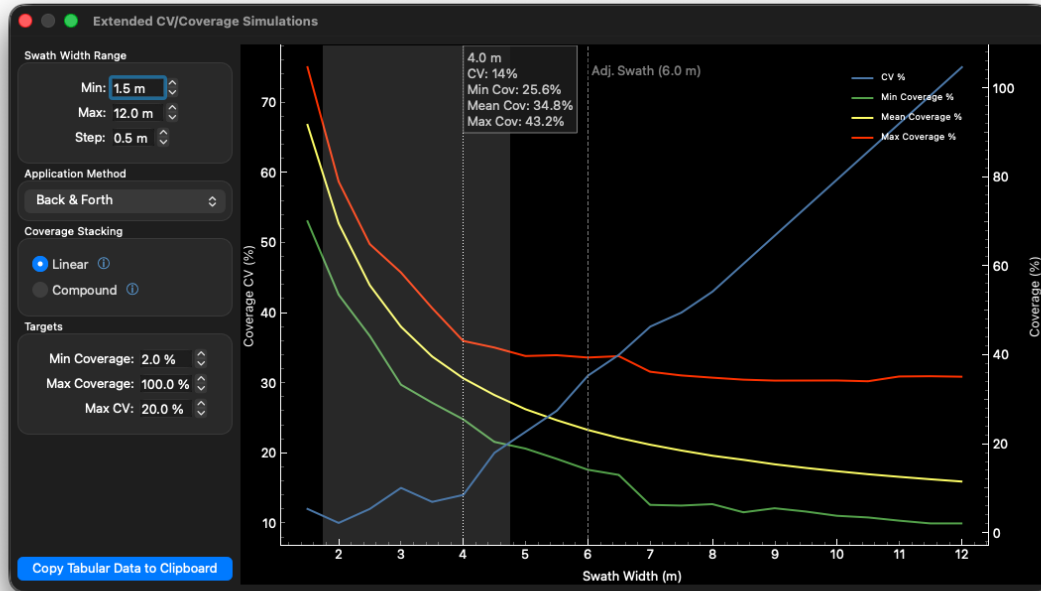


Figure 32: Spray-Card Analysis - Simulations - Extended

8.5 Droplet Distribution Tab

The **Droplet Distribution** sub-tab contains two histogram plots (Figure 33):

- **Spray Volume Contribution:** [Upper Plot] After discretizing the droplet spectrum, shows how much relative spray volume is contained in droplets of each discrete bin.
- **Quantity:** [Lower Plot] After discretizing the droplet spectrum, shows how many droplets of each discrete bin are found.

A table to the right of the histogram plots contains some select labeled droplet spectrum statistics.

The droplet spectrum category provided by *AccuPatt* is based on reference nozzle laser diffraction measurements by USDA ARS. Because the measurement method is different the provided droplet spectrum category should not be considered absolute.

Dropdown menus at the upper-right facilitate selection of the scope of calculation. The first menu allows selection of a series-wise composite (disabling the second menu) or selection of a pass. The second menu allows selection of a pass-wise composite (for the above selected pass) or selection of an individual card. Cards and passes which have been actively excluded will not count toward any composite calculations.

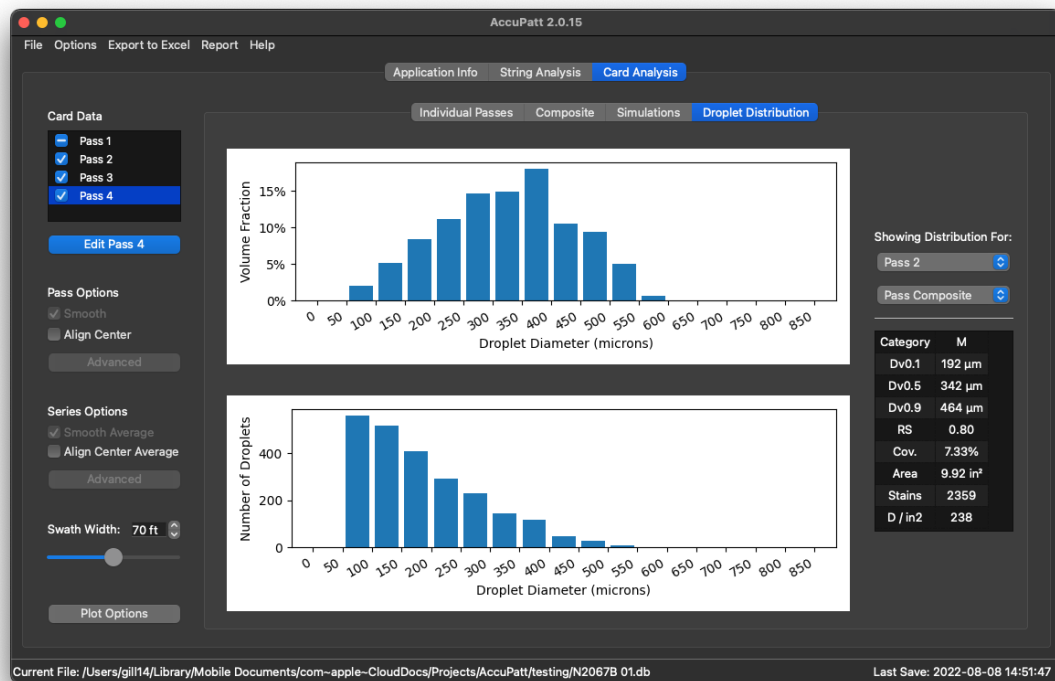


Figure 33: Spray-Card Analysis - Droplet Distribution

8.6 Pattern Processing Options

Pass Options affect only the selected pass:

- **Align Center:** Shift pattern left/right using chosen center method.

Series Options apply only to the average pattern.

- **Align Center Average:** Shift pattern left/right using chosen center method.

Swath Width defaults to the target swath width but is saved independently as the **Adjusted Swath Width**. This is the value that drives the average overlay (if inner-swath-half-average is selected), simulations and CV calculations.

The **Plot Options** button opens a popup with alternative, but equivalent, control to those in [section 4.4.7](#).

9 Operation SAFE Report

Creates a standardized printable/distributable PDF report for both string and spray card data. It can be configured/customized as discussed in [section 4.5](#).

String Summary Page (Figure 34) Headers with all reported application info and observables. Plots overlay, average and simulations. Includes a uniformity (Coefficient of Variation) table. Page will only be included if at least one string pass is collected and selected for inclusion in the composite.

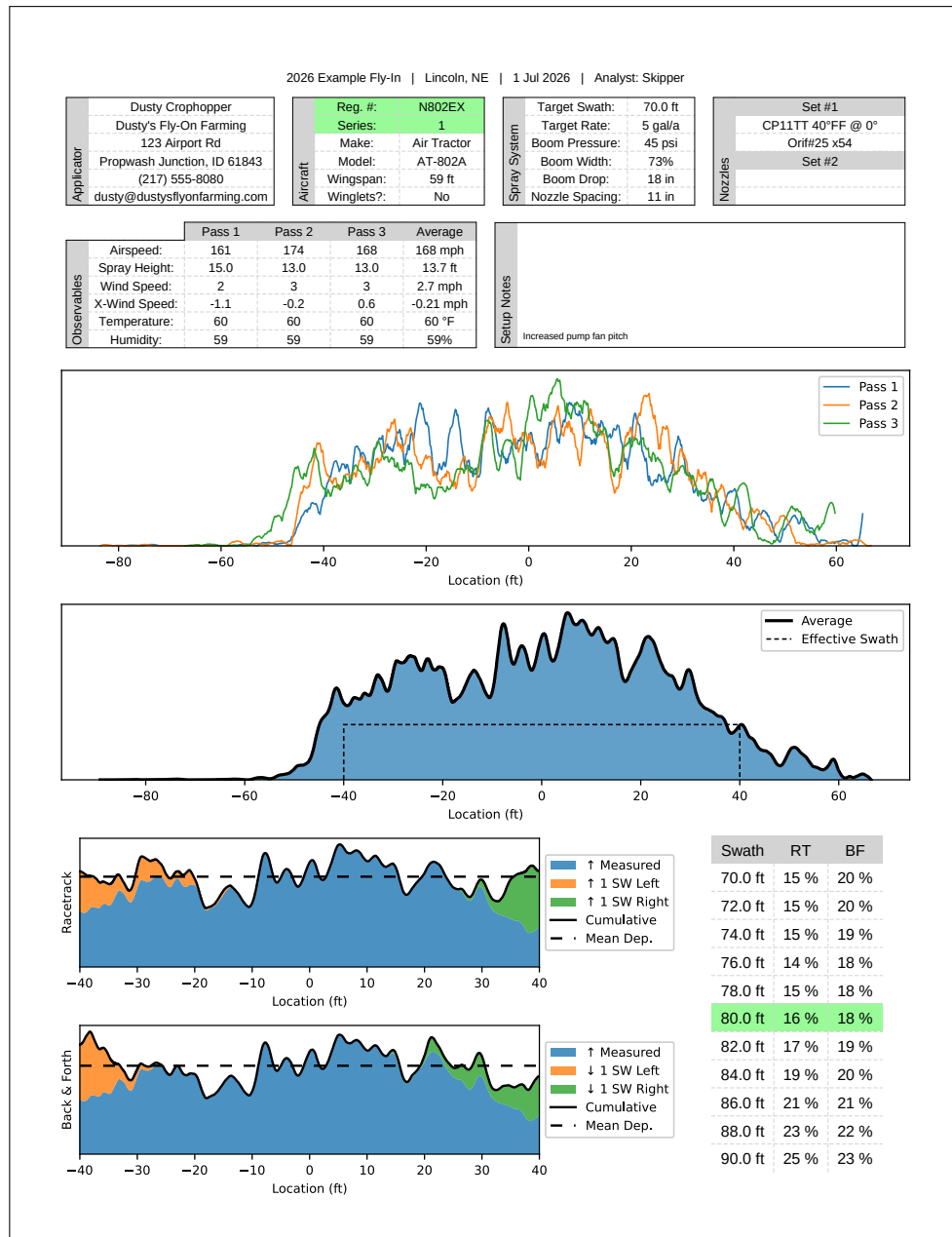


Figure 34: Operation SAFE Report - String Summary Page

Card Summary Page (Figure 35) Headers with all reported application info and observables. Plots coverage and droplet spectrum category by location, spray volume by droplet diameter and number of droplets by droplet diameter. Pass-wise droplet spectrum characteristics are shown for measured data as well as a USDA Model simulation for the provided nozzle/observable parameters. One page will be included for each pass which has card data associated with it and is selected for inclusion in the composite.

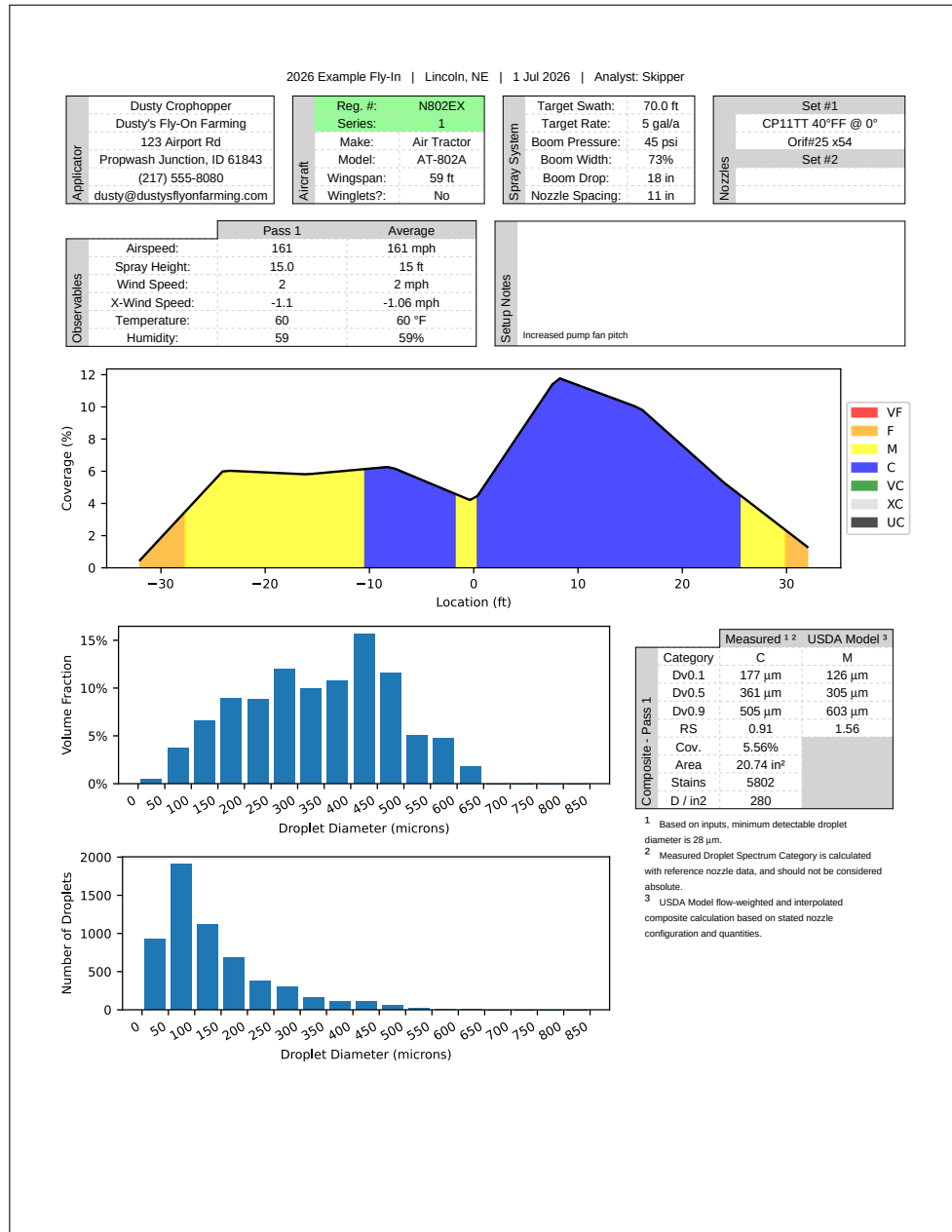


Figure 35: Operation SAFE Report - Card Summary Page

Card Image Page (Figure 36) Individual card images displayed as per the report settings. Individual card statistics are shown below each card. Pages are included for each pass which has card data associated with it and is selected for inclusion in the composite. Number of pages required will vary based on number of cards and report options selected.

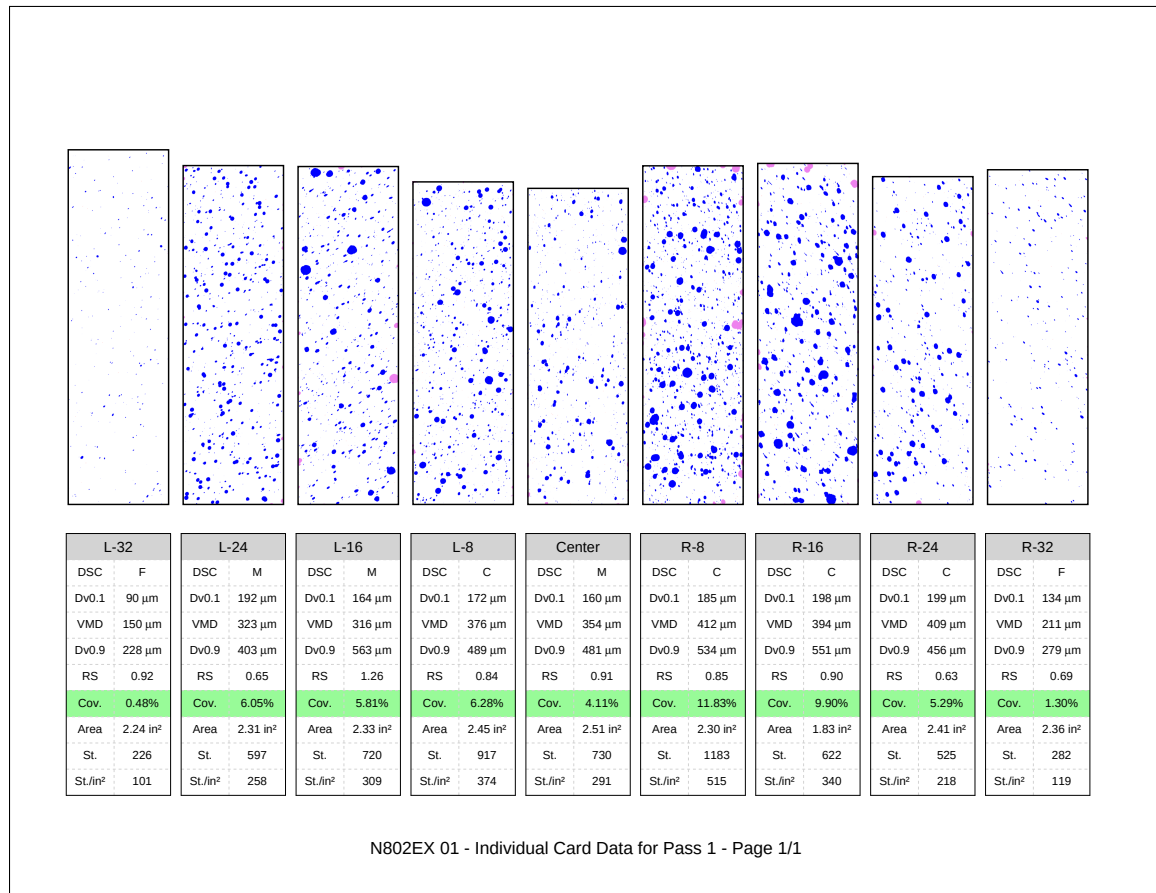


Figure 36: Operation SAFE Report - Card Image Page

10 Exporting Data to Excel

While typically unnecessary, it is sometimes desirable to export data obtained in *AccuPatt* into a format more widely accessible. To facilitate this, data can be copied to a *Microsoft Excel* (*.xlsx) file by choosing one of the following options from the **Export** menu:

- **Operation S.A.F.E. Attendee Log:** Generates the file required by NAAREF to document attendance at a S.A.F.E. Fly-In. Can choose whether to select desired files or to select a directory (folder) from which all AccuPatt (*.db), AccuPatt Legacy (*.xlsx), USDA (*.txt) and WRK (*.A) files will be included. This report is fully parameterized, so that it may be generated across multiple fly-ins and/or generating software types.
- **All Raw Data:** Exactly like it sounds, no processing or calculations, just the data and parameter settings.
- **Processed Data:** To-Do
- **All Images:** To-Do

Datafiles are stored as self-contained SQLite3 databases which can be opened directly using a tool such as [DB Browser for SQLite](#). See [section 14](#) for schema.

11 Opening Past Datafiles

From the menubar choose **File** → **Open** to open a filechooser window and select the file to open. When opening AccuPatt Legacy (*.xlsx), USDA (*.txt) or WRK (*.A) will cause a choice prompt to open the file as **View Only** or to **Create Compatible Copy** (Figure 37). If the latter is chosen, a native file will be created ([series.registration] [series.series-number].db) and placed in the same directory as the original file. Regardless of choice, the original file is never edited, only read.

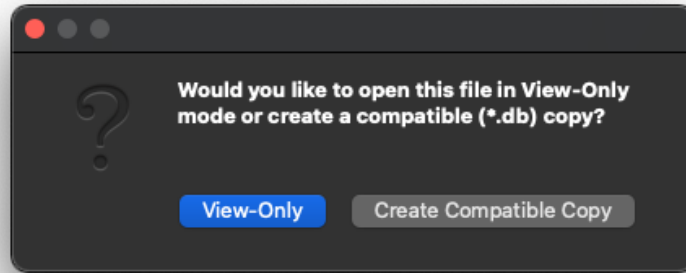


Figure 37: Open Datafile

12 Database Browser

It is often necessary to search a repository of data files for some specific parameters, such as registration number, aircraft model or nozzle to either find a specific series or view all matching series for broader analysis.

Clicking **File** → **Database Browser** will open an interactive search for this purpose (Figure 38).

Reg #	Series #	Make	Model	Target Swath	Target Rate	Type	Size	Deflection	Quantity	Adj. Swath (String)	RT CV	B&F CV	Modeled DSC
N802RH	12	Air Tractor	AT-802A	75.0 ft	2.4 gal/a	CP11TT 20*FF	12	0	42	70.0 ft	20%	18%	M
N802MB	1	Air Tractor	AT-802A	80.0 ft	5.0 gal/a					80.0 ft	21%	21%	
N8521Q	4	Air Tractor	AT-502B	65.0 ft	5.0 gal/a	CP11TT 40*FF	25	0	37	65.0 ft	20%	14%	C
N802EX	1	Air Tractor	AT-802A	70.0 ft	5.0 gal/a	CP11TT 40*FF	25	0	54	80.0 ft	16%	18%	M
1	1	Air Tractor	AT-602	80 ft	2.0 gal/a	CP09	0.078	5	39	75 ft	18%	17%	F
C-GMQA	1	Airbus	AS350B3e	16 m	75.0 l/ha					16 m	19%	19%	
1		Grumman	G-164A Ag Cat	40 ft						40 ft			

Figure 38: Interactive Database Browser

After choosing a directory, all **.db* files within it are recursively obtained and a row created in the table for its series. A chosen subset of columns are shown by default, but these can be optionally hidden and or additional columns shown via the **Columns** dropdown.

A global filter line allows filtering to all rows with any column containing the entered text. Each column header can be left clicked to sort by that column or right clicked to set a column specific filter (all column filters are additive).

The table can be exported to Excel in its current format by clicking **Export to Excel**.

Selecting any individual row will enable two additional buttons. **Reveal in Finder...** will open the file's parent folder on your file system. This is useful if you store PDF reports or accompanying photos in that folder. **Open in AccuPatt** will do just that, opening the datafile into AccuPatt for further analysis.

The table may be somewhat slow to load for large repositories (≈ 30 files per second) due to each series being fully processed for string CV calculations and nozzle-based DSC model projections.

13 Atomization Model & Calibration Calculator

Clicking **Extras** → **Atomization Model** will open the **Atomization Model** window. This is a dual-purpose interface to the USDA atomization models and calibration calculator ([Figure 39](#)). This tool is useful to quickly and simultaneously assess the projected impacts of nozzle set modifications on required flow and droplet spectrum.

The screenshot shows the 'Atomization Model' window with the following data:

Step	Field	Value	Range/Options
Step 1: Nozzle Set 1	Nozzle	CP11TT SS	
	Orifice	8	6 to 25
	Nozzle Angle	0	0, 15, 30, 45
	Quantity	29	
Step 2: Nozzle Set 2 (optional)	Nozzle	CP11TT SS	
	Orifice	6	6 to 25
	Nozzle Angle	0	0, 15, 30, 45
	Quantity	30	
Step 3: Operating Conditions	Pressure (psi)	55	30 to 90 psi
	Airspeed (mph)	170	70 to 180 mph
Step 4: Calculate Boom Flow Required (optional)	Application Rate (GPA)	2	
	Swath Width (ft)	70	
Output	DV0.1 (μm)	149 - M	
	DV0.5 (μm)	346 - M	
	DV0.9 (μm)	654 - C	
	DSC	M	
	Relative Span	1.460	
	%V<100 μm	5	
%V<200 μm		22	
Total Flow Rate (gpm)		48.193 (100.2% of req'd)	

Figure 39: Interactive Atomization Model and Calibration Calculator

If the currently open series has nozzles, pressure and airspeed (calculated from ground speed, wind speed, wind direction and pass heading) entered, these fields will pre-populate.

Entering in Nozzles (Step 1 and optionally Step 2) and Operating Conditions (Step 3) will output the model-projected droplet spectrum as well as total flow rate.

If the currently open series has application rate and swath width entered, these fields will also pre-populate (Step 4). If they are populated, the required flow rate is computed and the output will also compare the calculated flow rate to the required flow rate and show the required flow rate as a percentage of the required flow rate.

14 Data Structure

To the greatest extent practical, all provided analysis options are persistent for a saved data file. This is to ensure that, when re-opened at a later date or on another computer, results will be as close to identical as possible. Datafiles are stored as self-contained SQLite3 databases which can be opened directly using a tool such as [DB Browser for SQLite](#).

Below are the tables and relational outline for a datafile.

alembic_version

Variable	Type	Expected Values
version_num*	VARCHAR(32)	Unique, Maintains version compatibility

series

Variable	Type	Expected Values
id*	TEXT	Unique, Autogenerated
series	INTEGER	1, 2, 3, etc.
created	INTEGER	Unix Timestamp
modified	INTEGER	Unix Timestamp
notes_setup	TEXT	
notes_analyst	TEXT	
version_major	INTEGER	
version_minor	INTEGER	
version_release	INTEGER	

flyin

Variable	Type	Expected Values
series_id*	TEXT	
flyin_name	TEXT	
flyin_location	TEXT	
flyin_date	INTEGER	Unix Timestamp
flyin_analyst	TEXT	

applicator

Variable	Type	Expected Values
series_id*	TEXT	
pilot	TEXT	
business	TEXT	
street	TEXT	
city	TEXT	
state	TEXT	
zip	TEXT	
phone	TEXT	
email	TEXT	

aircraft

Variable	Type	Expected Values
series_id*	TEXT	
regnum	TEXT	
make	TEXT	
model	TEXT	
wingspan	REAL	
wingspan_units	TEXT	ft, m
winglets	TEXT	Yes, No

spray_system

Variable	Type	Expected Values
series_id*	TEXT	
swath	REAL	
swath_units	TEXT	ft, m
rate	REAL	
rate_units	TEXT	gpa, l/ha
pressure	REAL	
pressure_units	TEXT	psi, bar, kpa
boom_width	REAL	
boom_width_units	TEXT	ft, m
boom_drop	REAL	
boom_drop_units	TEXT	in, cm
nozzle_spacing	REAL	
nozzle_spacing_units	TEXT	in, cm

nozzles

One row per set of nozzles.

Variable	Type	Expected Values
id*	TEXT	1, 2, 3, etc.
series_id	TEXT	
type	TEXT	
size	TEXT	
deflection	TEXT	
quantity	INTEGER	

series_string

Variable	Type	Expected Values
series_id*	TEXT	
average_center	INTEGER	0, 1
average_center_method	TEXT	centroid, center-of-distribution
average_smooth	INTEGER	0, 1
average_smooth_window	REAL	
average_smooth_order	INTEGER	
equalize_integrals	INTEGER	0, 1
swath_adjusted	REAL	
simulated_adjascent_passes	INTEGER	No longer used

series_spray_card

Variable	Type	Expected Values
series_id*	TEXT	
average_center	INTEGER	0, 1
average_center_method	TEXT	centroid, center-of-distribution
swath_adjusted	REAL	
simulated_adjascent_passes	INTEGER	No longer used

passes

One row per pass.

Variable	Type	Expected Values
id*	TEXT	Unique, Autogenerated
series_id	TEXT	
pass_name	TEXT	
pass_number	INTEGER	
ground_speed	REAL	
ground_speed_units	TEXT	mph, kph, kn
spray_height	REAL	
spray_height_units	TEXT	ft, m
pass_heading	INTEGER	0-359
wind_direction	INTEGER	0-359
wind_speed	REAL	
wind_speed_units	mph, kph, kn	
temperature	REAL	
temperature_units	TEXT	°F, °C
humidity	REAL	

pass_string

One row per pass.

Variable	Type	Expected Values
pass_id*	TEXT	
dye	TEXT	
excitation_wav	INTEGER	
emission_wav	INTEGER	
integration_time_ms	INTEGER	
boxcar_width	INTEGER	
trim_left	INTEGER	
trim_right	INTEGER	
trim_vertical	REAL	
rebase	INTEGER	0, 1
center	INTEGER	0, 1
center_method	TEXT	centroid, center-of-distribution
smooth	INTEGER	0, 1
smooth_window	REAL	
smooth_order	INTEGER	
data_loc_units	TEXT	ft, m
excitation_data	TEXT	JSON List[location, intensity]
emission_data	TEXT	JSON List[location, intensity]
include_in_composite	INTEGER	0, 1

pass_spray_card

One row per pass.

Variable	Type	Expected Values
pass_id*	TEXT	
center	INTEGER	0, 1
center_method	TEXT	centroid, center-of-distribution
include_in_composite	INTEGER	0, 1

spray_card

One row per spray card.

Variable	Type	Expected Values
id*	TEXT	Unique, Autogenerated
pass_id	TEXT	
name	TEXT	
location	REAL	
location_units	TEXT	ft, m
include_in_composite	INTEGER	0, 1
threshold_type	INTEGER	0 <i>grayscale</i> , 1 <i>HSB</i>
threshold_method_grayscale	INTEGER	0 <i>automatic</i> , 1 <i>manual</i>
threshold_grayscale	INTEGER	0-255
threshold_color_hue_min	INTEGER	0-255
threshold_color_hue_max	INTEGER	0-255
threshold_color_hue_pass	INTEGER	0 <i>band-reject</i> , 1 <i>band-pass</i>
threshold_color_saturation_min	INTEGER	0-255
threshold_color_saturation_max	INTEGER	0-255
threshold_color_saturation_pass	INTEGER	0 <i>band-reject</i> , 1 <i>band-pass</i>
threshold_color_brightness_min	INTEGER	0-255
threshold_color_brightness_max	INTEGER	0-255
threshold_color_brightness_pass	INTEGER	0 <i>band-reject</i> , 1 <i>band-pass</i>
watershed	INTEGER	0, 1
min_stain_area_px	INTEGER	
stain_approximation_method	TEXT	none, min-enclosing-circle, fit-ellipse, convex-hull
dpi	INTEGER	
spread_method	INTEGER	0 <i>none</i> , 1 <i>direct</i> , 2 <i>adaptive</i> , 3 <i>logarithmic</i>
spread_factor_a	REAL	
spread_factor_b	REAL	
spread_factor_c	REAL	
has_image	INTEGER	0, 1
image	BLOB	